

Engineering Chemistry-1

CHE109

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Atoms rarely occurs in the free state at RT

Atoms of most of the elements have tendency to combine

A chemical bond is defined as the attractive force that holds two or more atoms together in a molecule or in an ions.

Why and How atoms combine?

Why do atoms combine?

- ❖ Net attractive forces between atoms

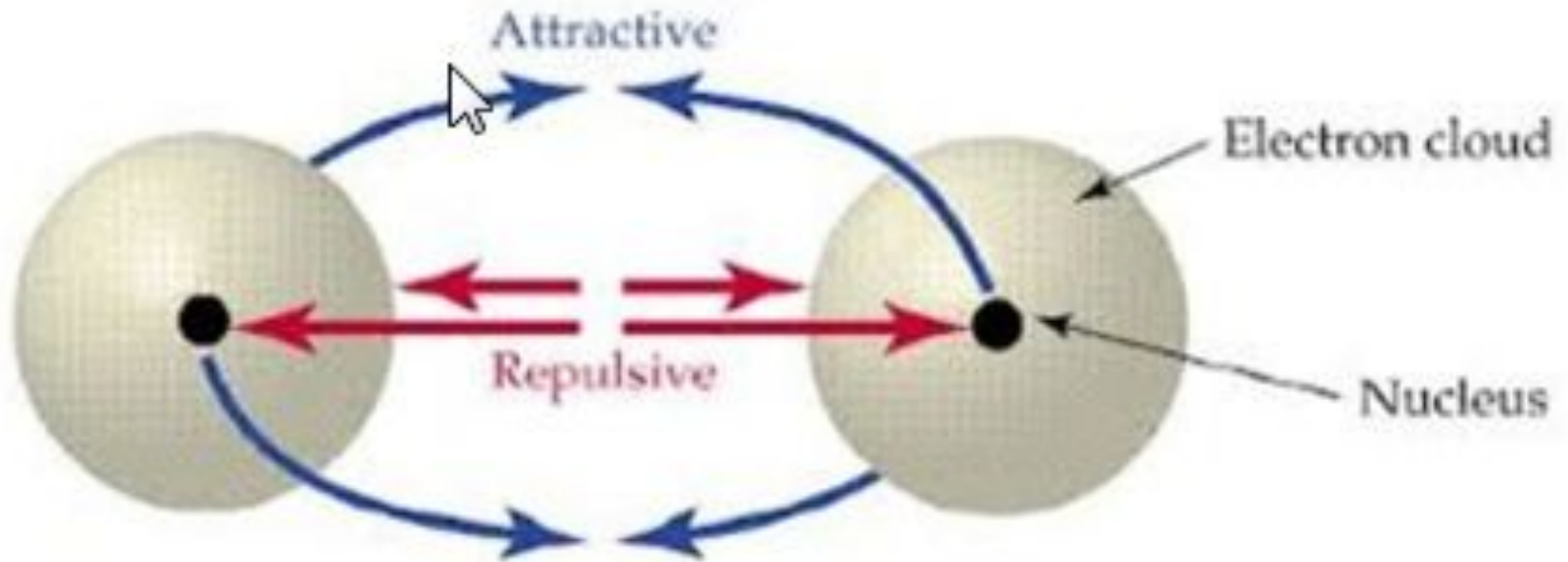


Fig. attractive and repulsive forces operating between two atoms when they come closer.

Why do atoms combine?

❖ Lowering of energy of combining atoms

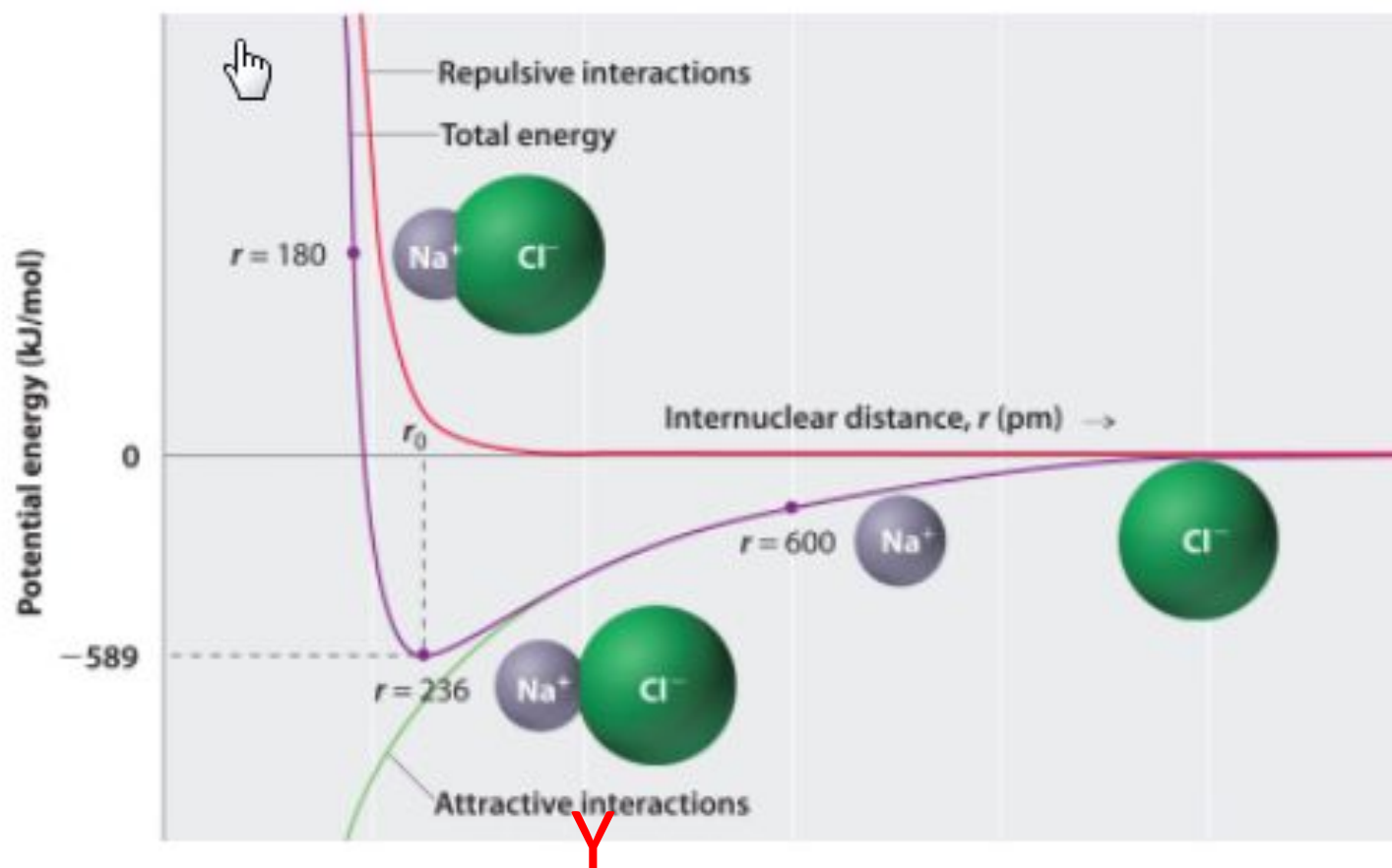


Fig. A plot of potential energy versus inter-nuclear distance for the interaction between a gaseous Na^+ ion and a gaseous Cl^- Ion

Chemical Bond

A chemical bond is defined as a force that acts between two or more atoms to hold them together as a stable molecule.

As we will study later, there are three different types of bonds recognized by chemists :

- (1) Ionic or Electrovalent bond
- (2) Covalent bond
- (3) Metallic bond

Valence

The term valence (or valency) is often used to state the potential or capacity of an element to combine with other elements. At one time, it was useful to define valence of an element as : the number of hydrogen atoms or twice the number of oxygen atoms with which that element could combine in a binary compound (containing two different elements only).

Valence is the number of bonds formed by an atom in a molecule.

Valence Electrons

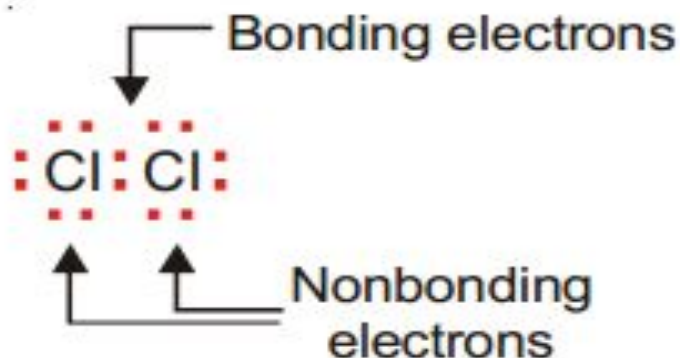
The electrons in the outer energy level of an atom are the ones that can take part in chemical bonding. These electrons are, therefore, referred to as the valence electrons.

The electronic configuration of Na is 2, 8, 1 and that of Cl is 2, 8, 7. Thus sodium has one valence electron and chlorine 7. It is important to remember that for an A group element of the periodic table (H, O, K, F, Al etc.) the group number is equal to the number of valence electrons.

Bonding and Non-bonding Electrons

The valence electrons actually involved in bond formation are called bonding electrons. The remaining valence electrons still available for bond formation are referred to as non-bonding electrons.

Thus :



Lewis Symbols of Elements

A Lewis symbol of an element consists of an element's symbol and surrounding dots to represent the number of valence electrons. In this notation, the symbol of an element represents the nucleus plus the inner normally filled levels (or shells) of the atom. For illustration, the symbol Na stands for the nucleus of sodium atom plus 2, 8 electrons in the inner two levels.

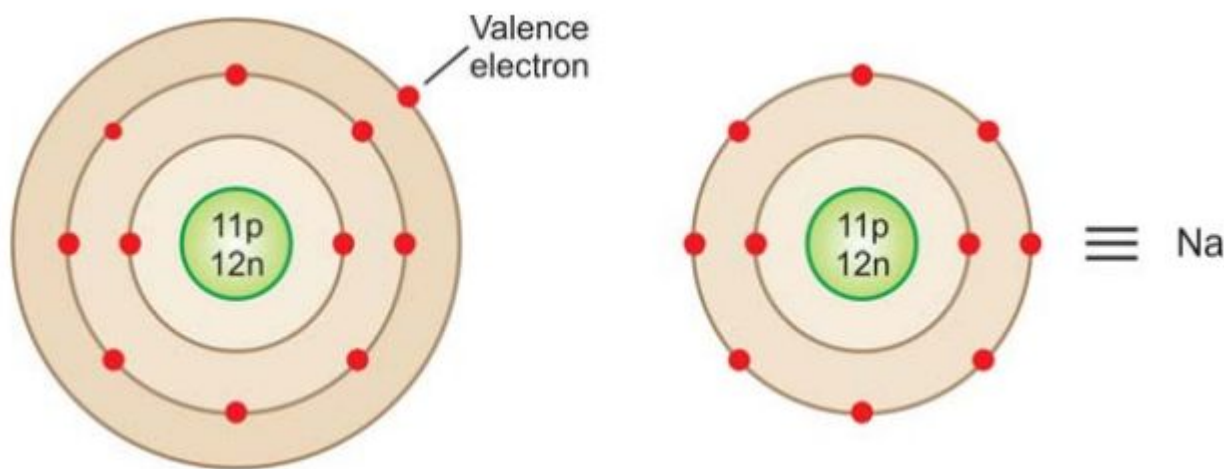


Fig. The Lewis symbol Na represents the nucleus and the electrons arranged in the inner two levels as 2, 8, minus the valence electrons.



The structural formulae of compounds built by union of Lewis symbols for the component atoms, are referred to as Electron-dot formulas, or Electron-dot structures or Lewis structures.

Electronic Theory of Valence

In 1916, they gave the electronic theory of valence. It states that : In chemical bond formation, atoms interact by losing, gaining, or sharing of electrons so as to acquire a stable noble gas configuration. Each noble gas, except helium, has a valence shell of eight electrons .

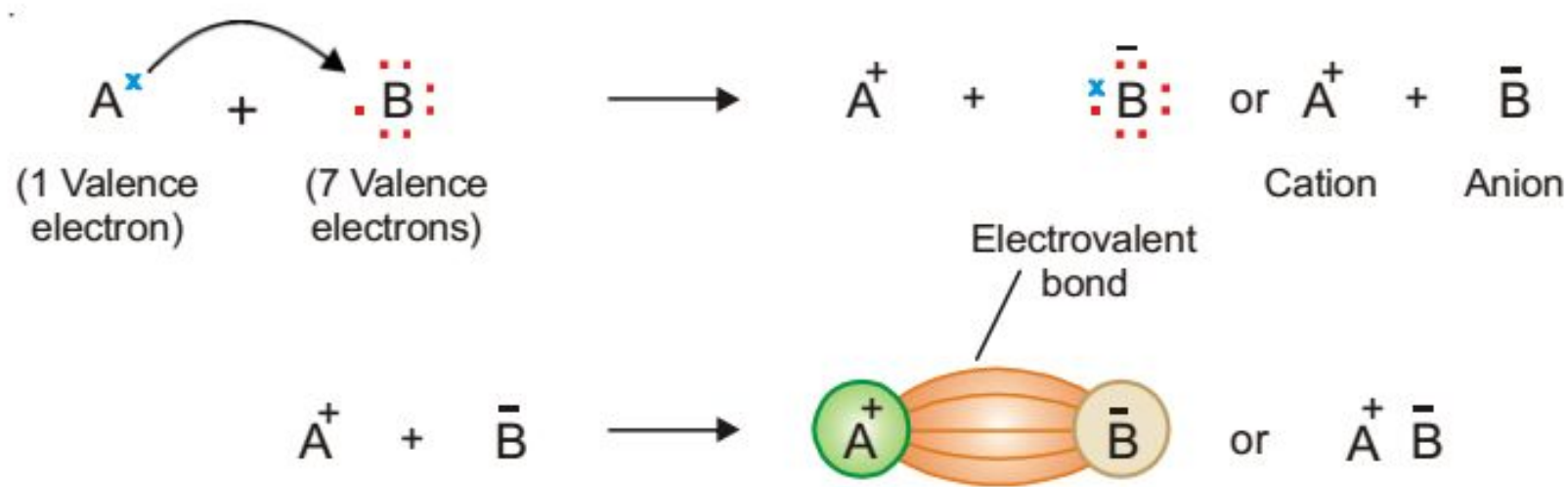
Noble gas	At. No.	Electrons in principal shells
He	2	2
Ne	10	2, 8
Ar	18	2, 8, 8
Kr	36	2, 8, 18, 8
Xe	54	2, 8, 18, 18, 8
Rn	86	2, 8, 18, 32, 18, 8

While atoms of noble gases possess a stable outer shell of eight electrons or octet, atoms of most other elements have incomplete octets. They may have less than 8 electrons or in excess. Thus, the electronic theory or valence could well be named as the **Octet theory of Valence**. It may be stated as : Atoms interact by electron-transfer or electron-sharing, so as to achieve the stable outer shell of eight electrons.

The tendency for atoms to have eight electrons in the outer shell is also known as the **Octet Rule or the Rule of Eight**. Since helium has two electrons in the outer shell, for hydrogen and lithium, having one and three (2, 1) electrons respectively, it is the **Rule of two** which will apply.

Ionic Bond

This type of bond is established by transfer of an electron from one atom to another. Let us consider a general case when an atom A has one electron in the valence shell and another atom B has seven electrons. A has one electron in excess and B has one electron short than the stable octet. Therefore, A transfers an electron to B and in this transaction both the atoms acquire a stable electron octet. The resulting positive ion (cation) and negative ion (anion) are held together by electrostatic attraction.



The electrostatic attraction between the cation (+) and anion (–) produced by electron-transfer constitutes an Ionic or Electrovalent bond.

The compounds containing such a bond are referred to as Ionic or Electrovalent Compounds.

Conditions For Formation Of Ionic Bond

- **Number of valence electrons**

The atom *A* should possess 1, 2 or 3 valence electrons, while the atom *B* should have 5, 6 or 7 valence electrons. The elements of group IA, IIA and IIIA satisfy this condition for atom *A* and those of groups VA, VIA, and VIIA satisfy this condition for atom *B*.

Net lowering of Energy

To form a stable ionic compound, there must be a net lowering of the energy.

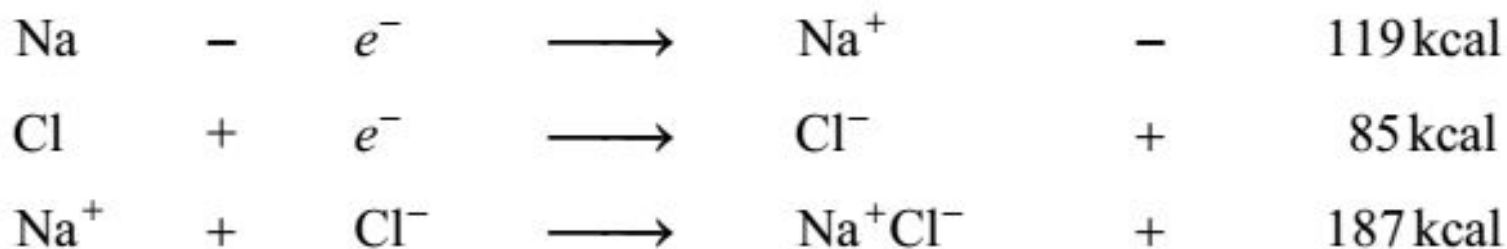
(a) The removal of electron from atom A ($A - e^- \rightarrow A^+$) requires input of energy, which is the ionization energy (IE). It should be low.

(b) The addition of an electron to B ($B + e^- \rightarrow B^-$) releases energy, which is the electron affinity of B (EA). It should be high.

(c) The electrostatic attraction between A^+ and B^- in the solid compound releases energy, which is the electrical energy. It should also be high.

If the energy released in steps (b) and (c) is greater than the energy consumed in step (a), the overall process of electron transfer and formation of ionic compound results in a net release of energy.

Therefore, ionisation of A will occur and the ionic bond will be formed. For example, in case of formation of sodium chloride (NaCl), we have



The net energy released is $187 + 85 - 119 = 153$ kcal. Since the overall process results in a lowering of energy, the ionic bond between Na and Cl will be formed.

- **Electronegativity difference of A and B**

From the line of argument used in (2), we can say that atoms A and B if they have greatly different electronegativities, only then they will form an ionic bond. In fact, a difference of 2 or more is necessary for the formation of an ionic bond between atoms A and B. Thus Na has electronegativity 0.9, while Cl has 3.0. Since the difference is $(3.0 - 0.9) = 2.1$, Na and Cl will form an ionic bond.

Factors Governing The Formation of Ionic Bond

(1) Ionisation Energy

Lower the ionisation energy greater will be the tendency of the metal atom of change into cation and hence greater will be the ease of formation of ionic bond. That is why alkali metals and alkaline earth metals form ionic bonds easily.

(2) Electron Affinity

The atom which accepts the electron and changes into anion should have high electron affinity. Higher the electron affinity more is the energy released and stable will be the anion formed.

(3) Lattice Energy

After the formation of cations and anions separately, they combine to form ionic compound.



In this process, energy is released. It is called Lattice Energy. It may be defined as “the amount of energy released when one mole of an ionic compound is formed from its cations and anions.” Greater the lattice energy, greater the strength of ionic bond. The value of lattice energy depends upon the following two factors :

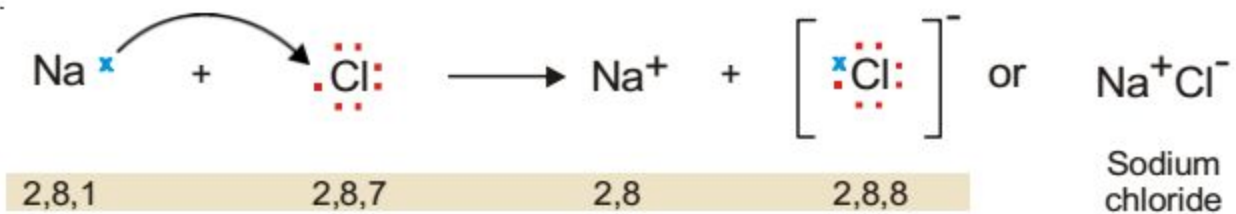
(a) Size of the ions

In order to have the greater force of attraction between the cations and anions their size should be small as the force of attraction is inversely proportional to the square of the distance between them.

(b) Charge on Ions

Greater the charge on ions greater will be the force of attraction between them and, therefore, greater will be the strength of the ionic bond.

Some Examples Of Ionic Compounds

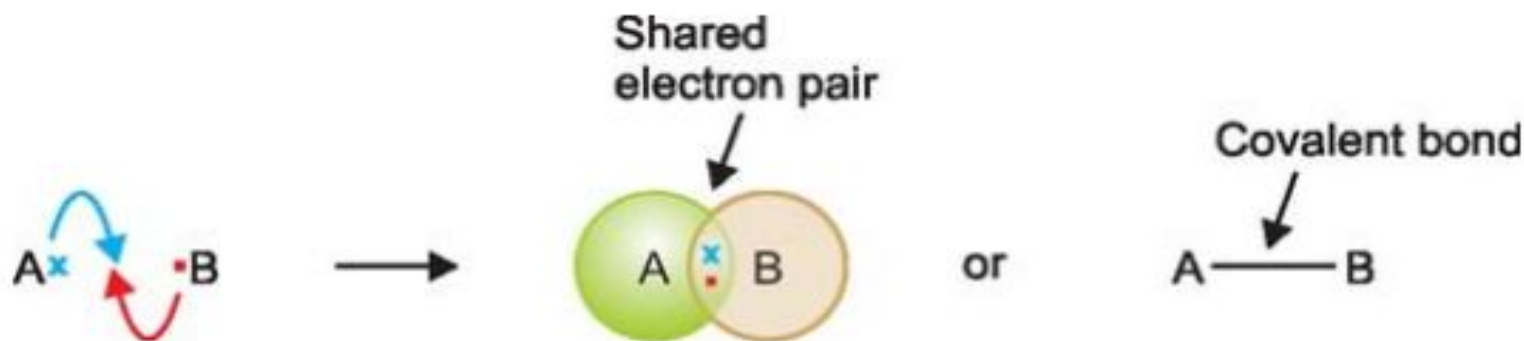


Characteristics of Ionic Compounds

1. Solids at Room Temperature
2. High Melting Points
3. Hard and brittle
4. Soluble in water
5. Conductors of electricity
6. Do not exhibit isomerism
7. Ionic reactions are fast

Covalent Bond

The electron transfer theory could not explain the bonding in molecules such as H_2 , O_2 , Cl_2 etc., and in organic molecules, that had no ions. It was G.N. Lewis who suggested that two atoms could achieve stable 2 or 8 electrons in the outer shell by sharing electrons between them. Let us consider a general case where an atom A has one valence electron and another atom B has seven valence electrons. As they approach each other, each atom contributes one electron and the resulting electron pair fills the outer shell of both the atoms. Thus A acquires stable 2 electrons and B, 8 electrons in the outer shell.



The shared pair is indicated by a dash (—) between the two bonded atoms. A shared pair of electrons constitutes a Covalent bond or Electron-pair bond.

Conditions For Formation of Covalent Bond

(1) Number of valence electrons

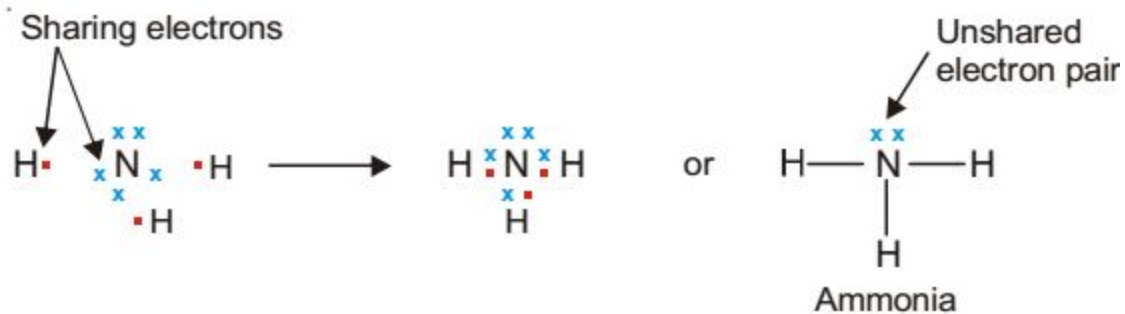
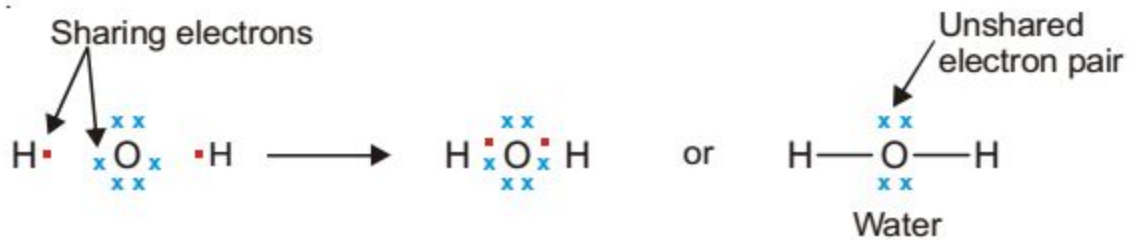
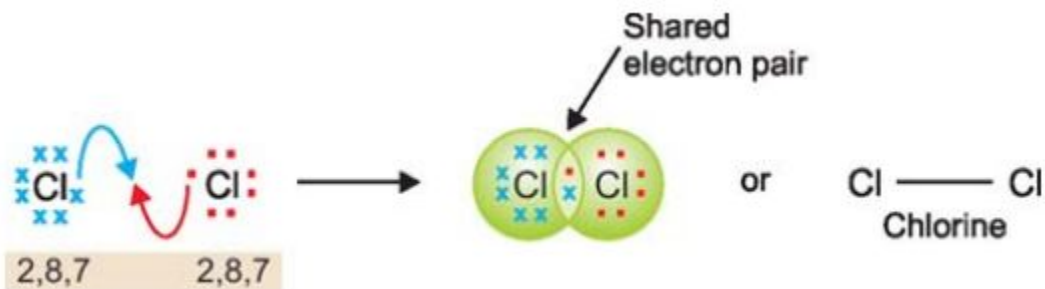
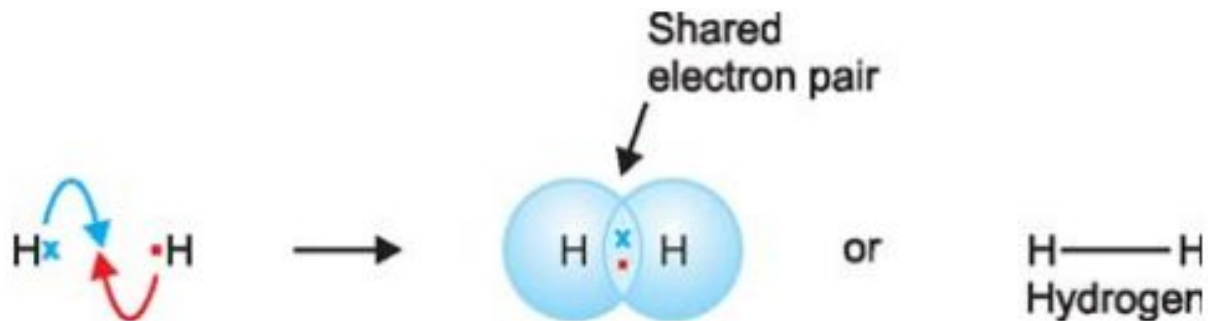
Each of the atoms A and B should have 5, 6 or 7 valence electrons so that both achieve the stable octet by sharing 3, 2 or 1 electron-pair. H has one electron in the valence shell and attains duplet. The non-metals of groups VA, VIA and VIIA respectively satisfy this condition.

(2) Equal electronegativity

The atom A will not transfer electrons to B if both have equal electronegativity, and hence electron sharing will take place. This can be strictly possible only if both the atoms are of the same element.

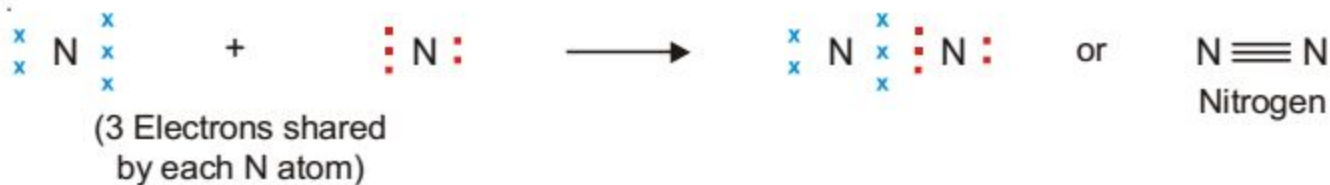
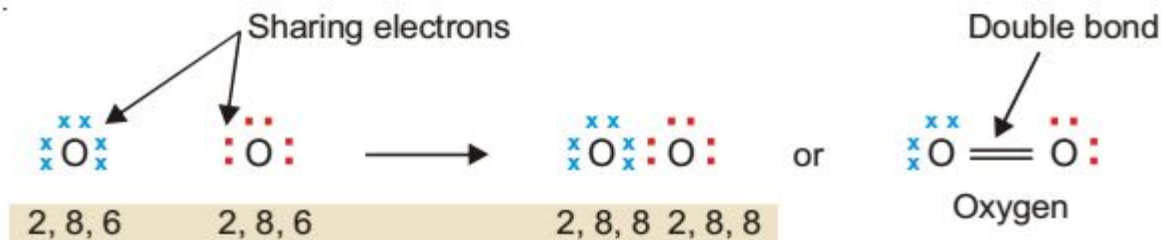
(3) Equal sharing of electrons

The atoms A and B should have equal (or nearly equal) electron affinity so that they attract the bonding electron pair equally. Thus equal sharing of electrons will form a nonpolar covalent bond. Of course, precisely equal sharing of electrons will not ordinarily occur except when atoms A and B are atoms of the same element, for no two elements have exactly the same electron affinity.



Examples Of Multiple Covalent Compounds

In many molecules, we find that in order to satisfy the octet, it becomes necessary for two atoms to share two or three pairs of electrons between the same two atoms. The sharing of two pairs of electrons is known as a Double bond and the sharing of three pairs of electrons a Triple bond. Let us consider some examples of compounds containing these multiple covalent bonds in their molecules.



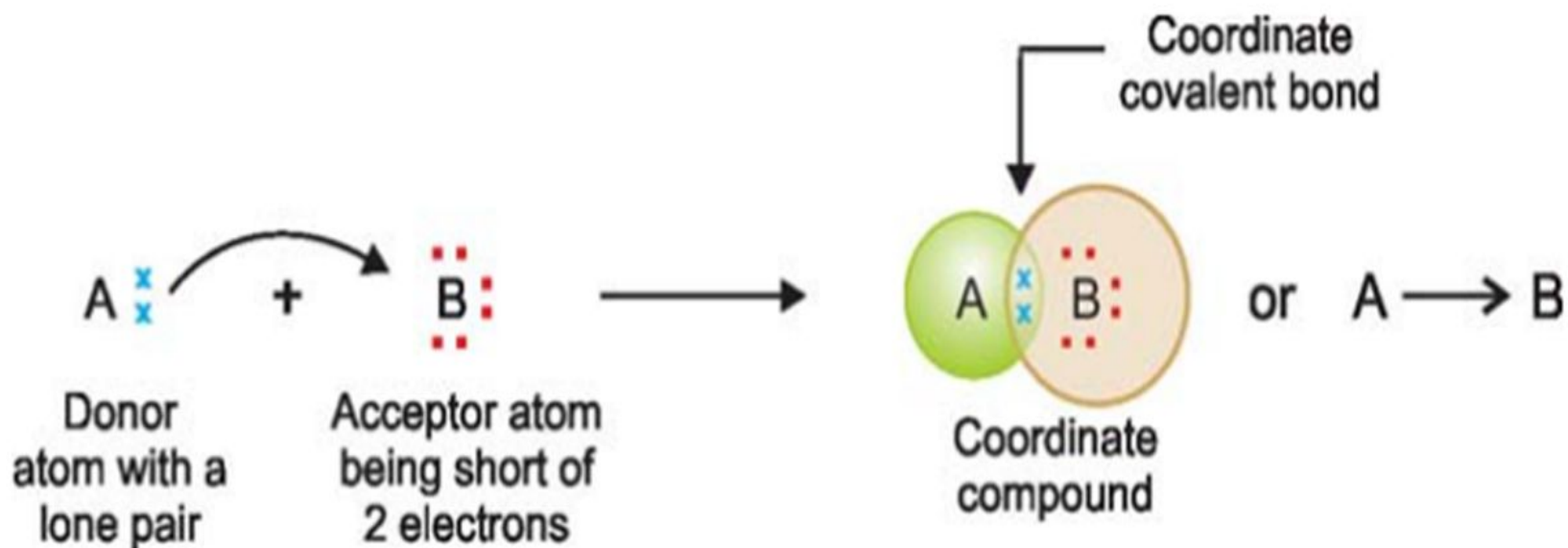
Characteristics of Covalent Compounds

- (1) Gases, liquids or solids at room temperature
- (2) Low melting points and boiling points
- (3) Neither hard nor brittle
- (4) Soluble in organic solvents
- (5) Non-conductors of electricity
- (6) Exhibit Isomerism
- (7) Molecular reactions

CO-ORDINATE COVALENT BOND

In a normal covalent bond, each of the two bonded atoms contributes one electron to make the shared pair. In some cases, a covalent bond is formed when both the electrons are supplied entirely by one atom. Such a bond is called co-ordinate covalent or dative bond. It may be defined as : a covalent bond in which both electrons of the shared pair come from one of the two atoms (or ions). The compounds containing a coordinate bond are called coordinate compounds.

If an atom A has an unshared pair of electrons (lone pair) and another atom B is short of two electrons than the stable number, coordinate bond is formed. A donates the lone pair to B which accepts it. Thus both A and B achieve the stable 2 or 8 electrons, the lone pair being held in common.



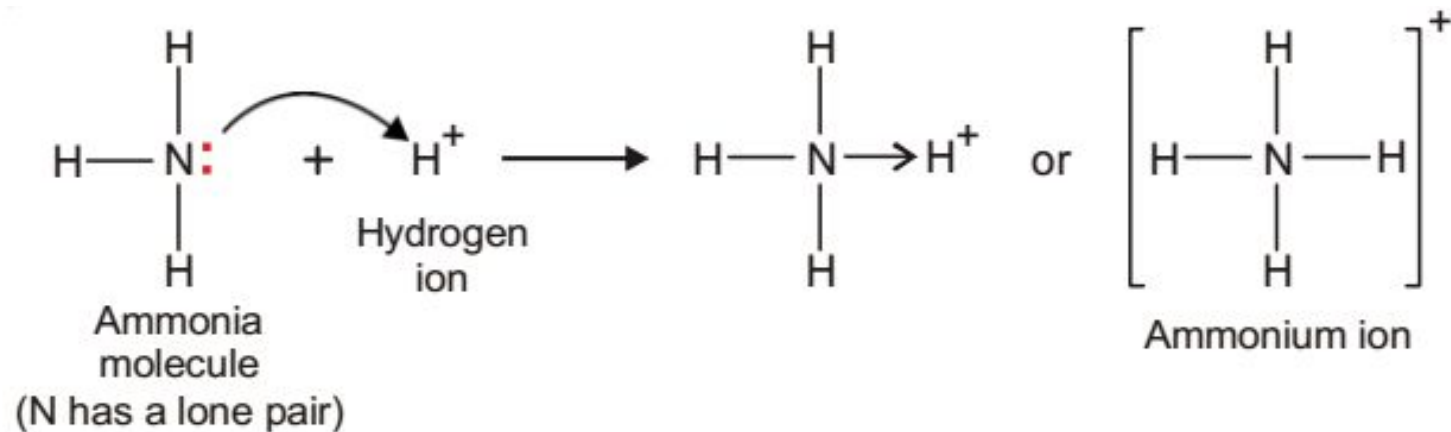
The atom A which donates the lone pair is called the donor, while B which accepts it the acceptor. The bond thus established is indicated by an arrow pointing from A to B. Although the arrow head indicates the origin of the electrons, once the coordinate bond is formed it is in no way different from an ordinary covalent bond.

The molecule or ion that contains the donor atom is called the ligand.

Some Examples of Coordinate Compounds or Ions

Ammonium ion, NH_4^+

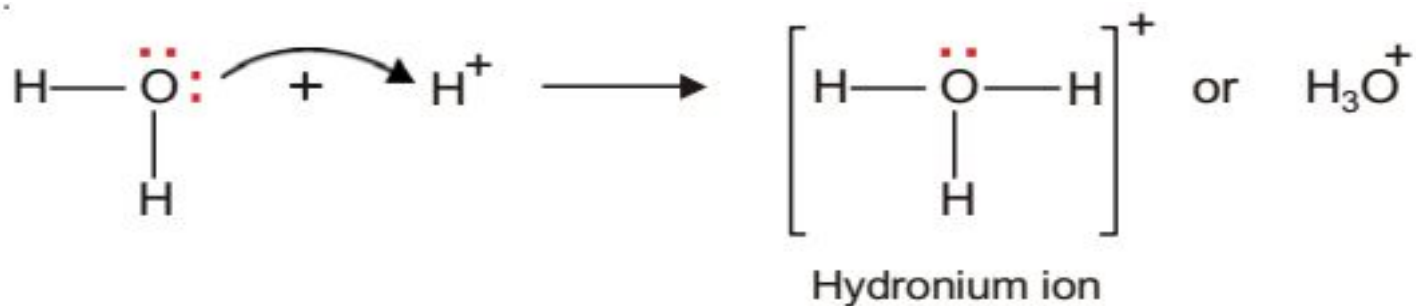
In ammonia molecule, the central N atom is linked to three H atoms and yet N has an unshared pair of electrons. The H^+ ion furnished by an acid has no electron to contribute and can accept a pair of electrons loaned by N atom. Thus, NH_3 donates its unshared electrons to H^+ forming ammonium ion.



All the N—H bonds in NH_4^+ are identical, once the coordinate bond $\text{N} \rightarrow \text{H}^+$ is established.

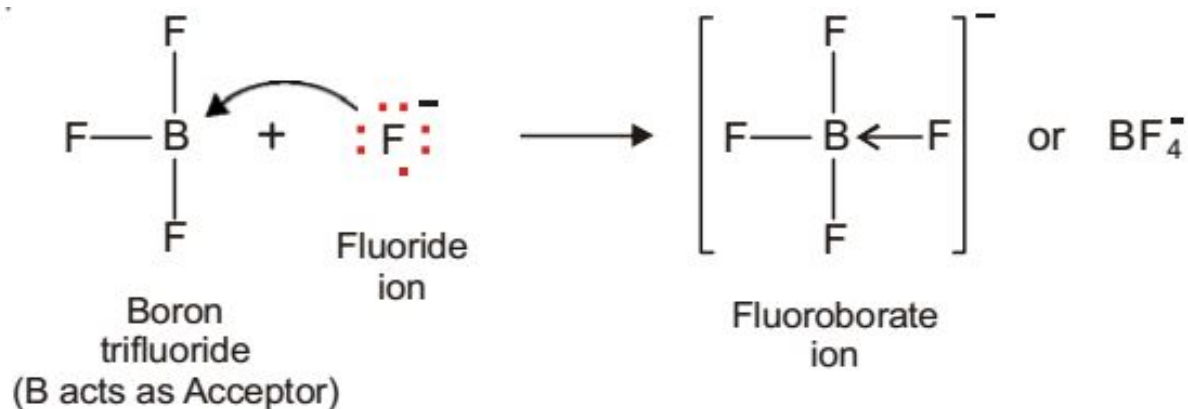
Hydronium ion, H_3O^+

The oxygen atom in water molecule is attached to two H atoms by two covalent bonds. There are still two unshared pairs of electrons with the O atom. The O atom donates one of these pairs of electrons to H^+ ion and the hydronium ion is thus formed.



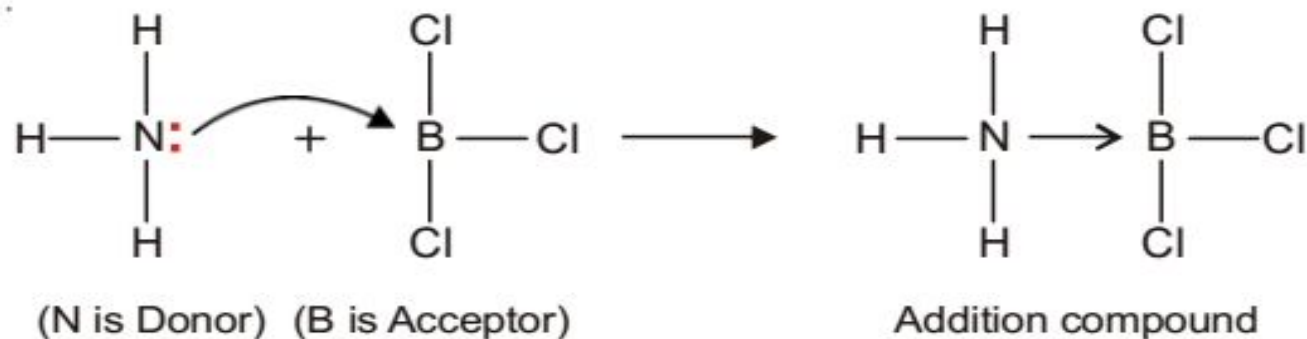
Fluoroborate ion, BF_4^-

It is formed when a boron trifluoride molecule (BF_3) shares a pair of electrons supplied by fluoride ion (F^-).



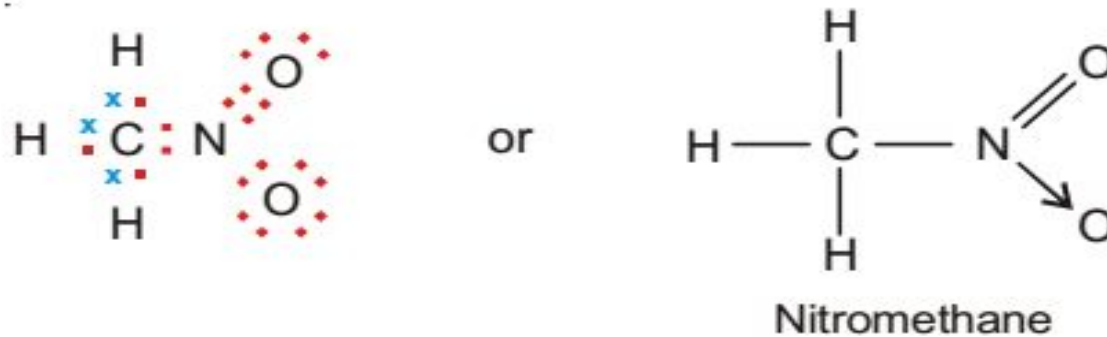
Addition compound of NH_3 with BCl_3

The N atom of ammonia molecule (NH_3) has lone pair while B atom in boron trichloride (BCl_3) is short of two electrons than stable octet. An addition compound is formed as the N atom donates its lone pair to B atom of BCl_3 .

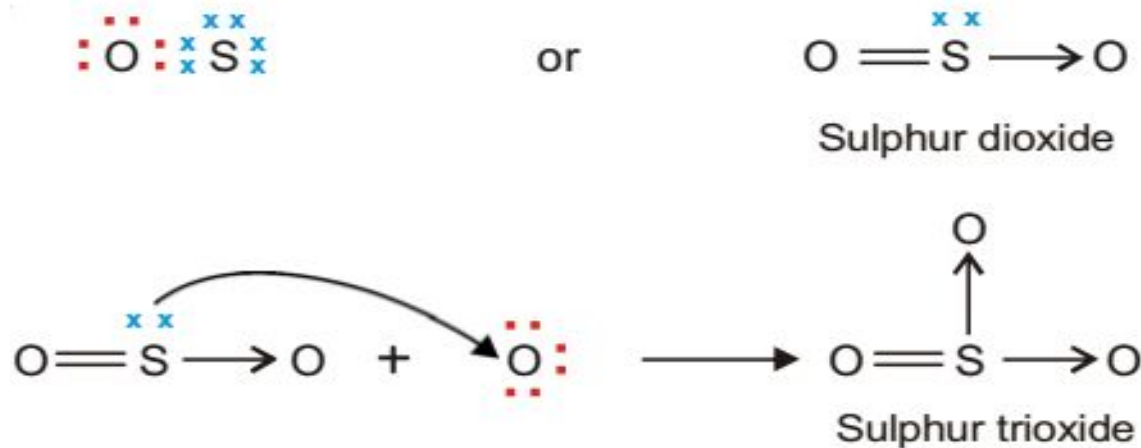


Nitromethane, CH₃NO₂

The Lewis structure of nitromethane is shown below. Here the N atom has five valence electrons, three of which are used in forming a covalent bond with C atom and two covalent bonds with O atom. The N atom is still left with two unshared electrons which are donated to another O atom.

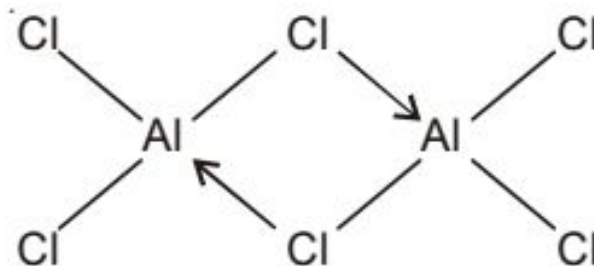


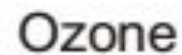
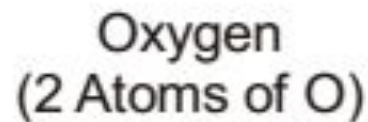
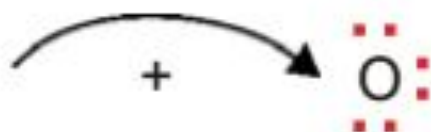
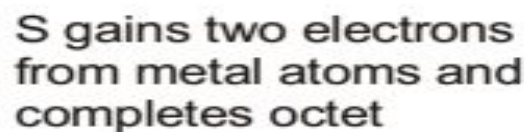
Sulphur dioxide, SO_2 , and Sulphur trioxide, SO_3 Sulphur achieves its octet by forming two covalent bonds with one O atom, giving SO. The S atom in SO has two lone pairs, one of which is shared with a second O atom to form sulphur dioxide, SO_2 . The S atom in SO_2 still has one lone pair which it donates to a third O atom forming the sulphur trioxide (SO_3) molecule.



Aluminium Chloride, Al_2Cl_6

Aluminium atom has three valence electrons which it shares with three Cl atoms, forming three covalent bonds. Thus the Al atom acquires six electrons in its outer shell. Now Cl atom has three lone pairs, one of which is donated to the Al atom of another molecule AlCl_3 . Thus both Al atoms achieve octet and stable Al_2Cl_6 results.





Ionic Bond

1. Formed by transfer of electrons from a metal to a non-metal atom.
2. Consists of electrostatic force between (+) and (−) ions.
3. Non-rigid and non-directional; cannot cause isomerism.

Properties of Compounds

1. Solids at room temperature.
2. High melting and boiling points.
3. Hard and brittle.
4. Soluble in water but insoluble in organic solvents.
5. Conductors of electricity
6. Undergo ionic reactions which are fast.

Covalent Bond

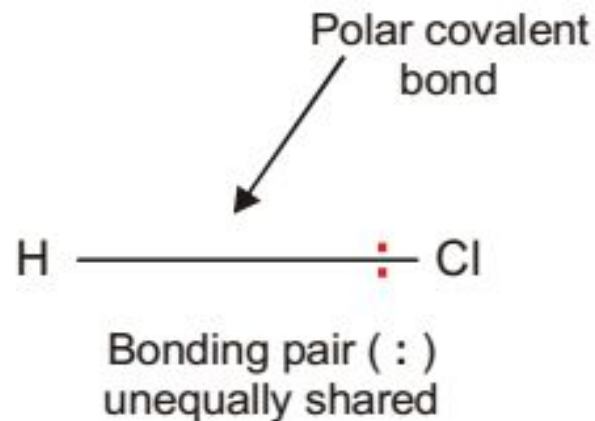
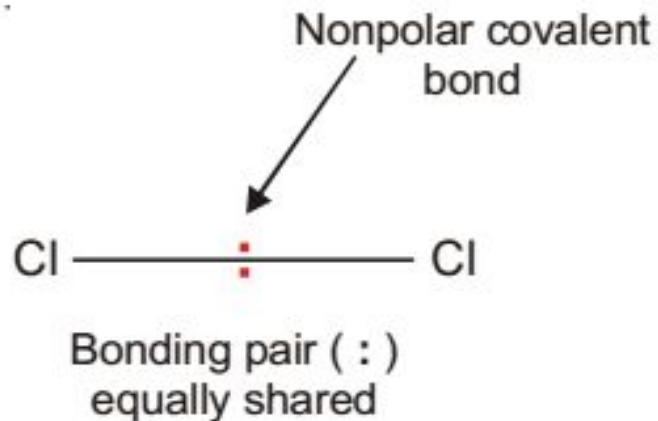
1. Formed by sharing of electrons between nonmetal atoms.
2. Consists of a shared pair of electrons between atoms.
3. Rigid and directional : causes stereoisomerism.

Properties of Compounds

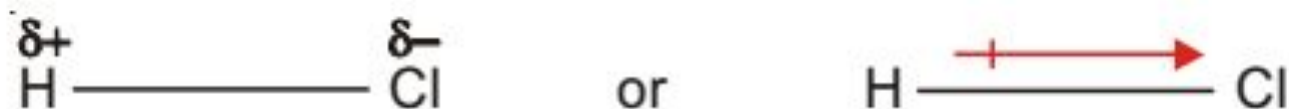
1. Gases, liquids or soft solids.
2. Low melting and boiling points.
3. Soft, much readily broken
4. Insoluble in water but soluble in organic solvents.
5. Non-conductors of electricity.
6. Undergo molecular reactions which are slow.

POLAR COVALENT BONDS

In the H_2 or Cl_2 molecule, the two electrons constituting the covalent bond are equally shared by the two identical nuclei. Due to even distribution of (+) and (−) charge, the two bonded atoms remain electrically neutral. Such a bond is called nonpolar covalent bond. However, when two different atoms are joined by a covalent bond as in HCl , the electron pair is not shared equally.



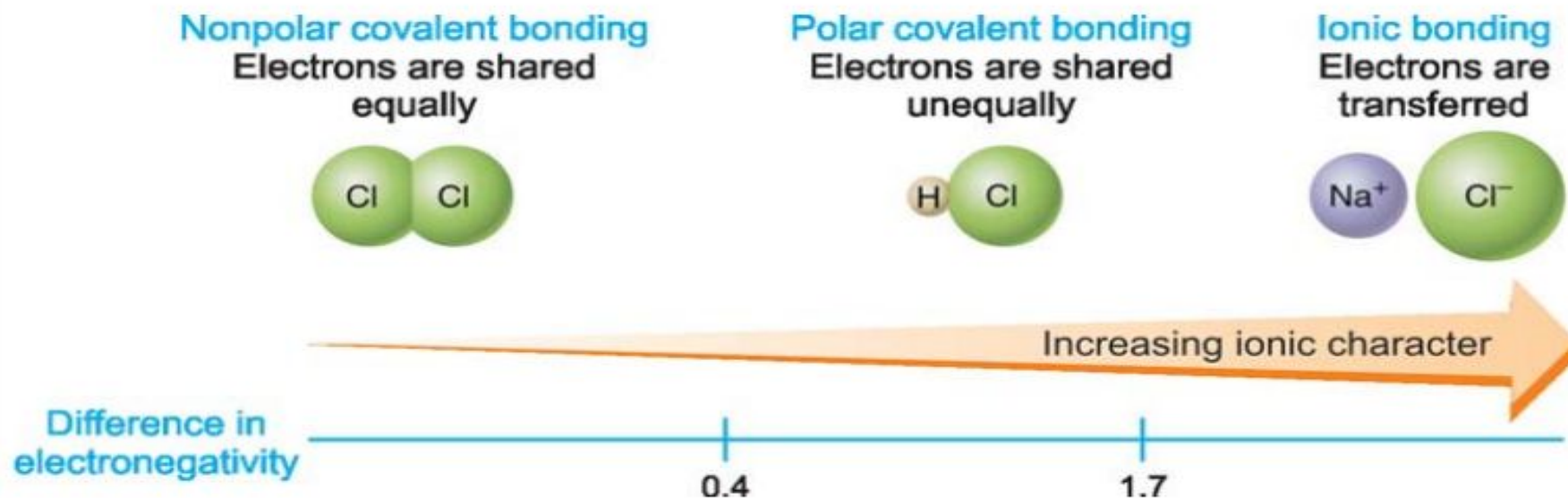
Due to a greater attraction of one nucleus (Cl) for the electrons, the shared pair is displaced towards it. This makes one end of the bond partially positive ($\delta+$) and the other partially negative ($\delta-$).



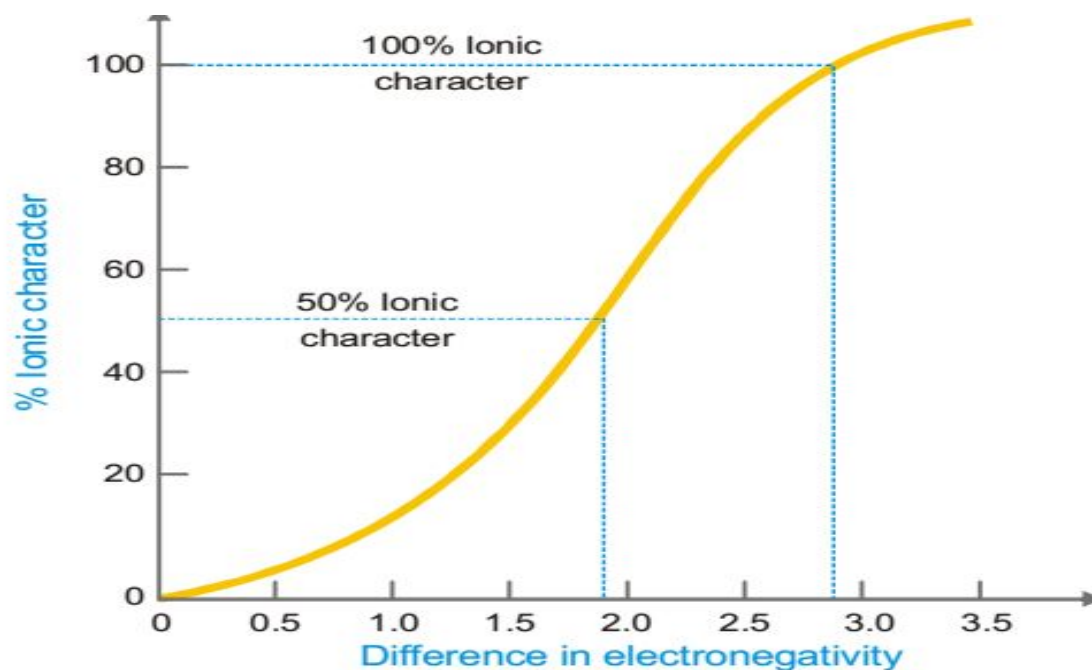
A covalent bond in which electrons are shared unequally and the bonded atoms acquire a partial positive and negative charge, is called a polar covalent bond.

A molecule having partial positive and negative charge separated by a distance is commonly referred to as a Dipole (two poles). The dipole of a bond is indicated by an arrow from positive to negative end with a crossed tail as shown above in HCl molecule.

the amount of polarity of a bond is determined by the difference of electronegativity (or tendency to attract electrons) of the two bonded atoms. The greater the difference of electronegativity between two atoms, greater the polarity. A graph showing the % age ionic character and difference in electronegativity between the two atoms is shown in Fig. 5.5.



As a matter of fact, if this difference is around 1.9 and 2.9, the bond is generally ionic, meaning that one atom has gained complete control of the electron pair in the bond. The percentage ionic character of a bond can be calculated by using the equation $\% \text{ ionic character} = 16 [X_A - X_B] + 3.5 [X_A - X_B]^2$. This equation was given by Hannay and Smith.



■ **Figure 5.5**
Graph between % ionic character and difference in electronegativity.

PROBLEM

Calculate the percentage ionic character of C–Cl bond in CCl_4 if the electronegativities of C and Cl are 3.5 and 3.0 respectively.

SOLUTION

$$\% \text{age ionic character} = 16 [X_A - X_B] + 3.5 [X_A - X_B]^2$$

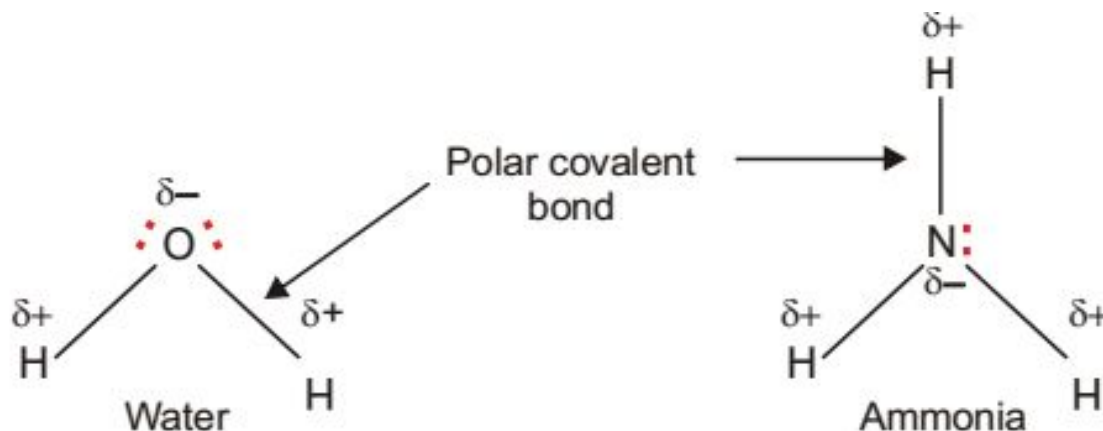
Give $X_A = 3.5$ and $X_B = 3.0$

$$\therefore \% \text{age ionic character} = 16 (3.5 - 3.0) + 3.5 (3.5 - 3.0)^2$$

$$= 8.0 + 0.875$$

$$= 8.875\%$$

Examples of Polar Covalent Bonds Water molecule (H_2O) contains two O–H covalent bonds. The electronegativity of O is 3.5 and that of H is 2.1. Thus both the bonds are polar and water has a polar molecule.



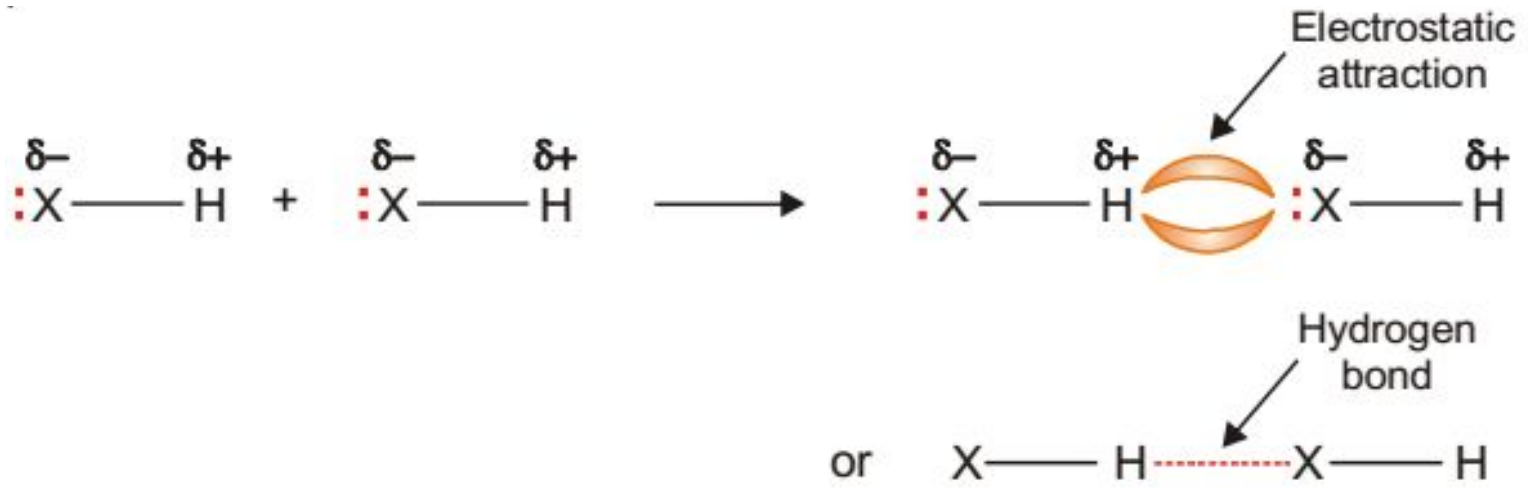
In ammonia molecule, there are three N–H bonds. The electronegativity of N is 3.0 and that of H is 2.1. Therefore all the N–H bonds are polar and ammonia has a polar molecule. The electronegativity of fluorine (F) is 4.0 and that of H is 2.1. The difference of electronegativities being very great, the molecule H–F has a strong dipole.

HYDROGEN BONDING (H-Bonding)

When hydrogen (H) is covalently bonded to a highly electronegative atom X (O, N, F), the shared electron pair is pulled so close to X that a strong dipole results



Since the shared pair is removed farthest from H atom, its nucleus (the proton) is practically exposed. The H atom at the positive end of a polar bond nearly stripped of its surrounding electrons, exerts a strong electrostatic attraction on the lone pair of electrons around X in a nearby molecule. Thus :



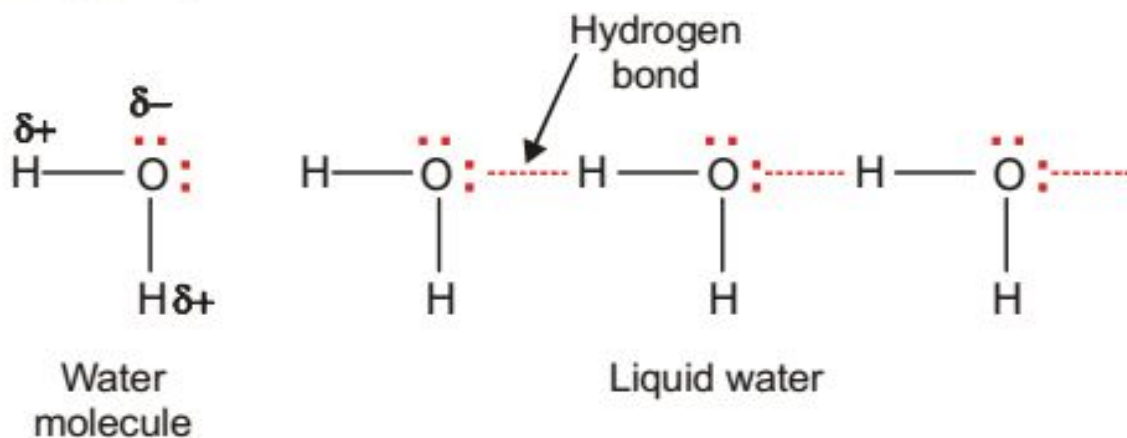
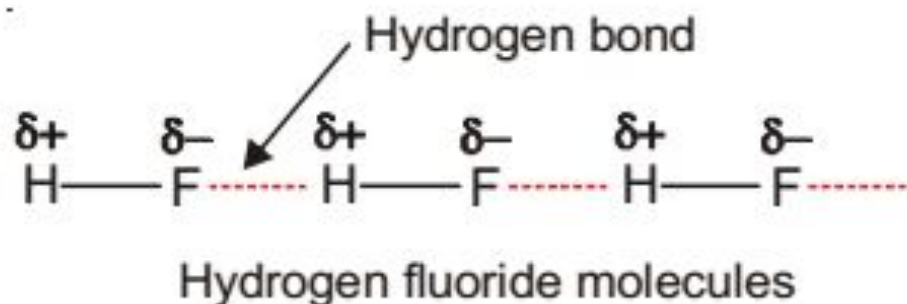
The electrostatic attraction between an H atom covalently bonded to a highly electronegative atom X and a lone pair of electrons of X in another molecule, is called Hydrogen Bonding.

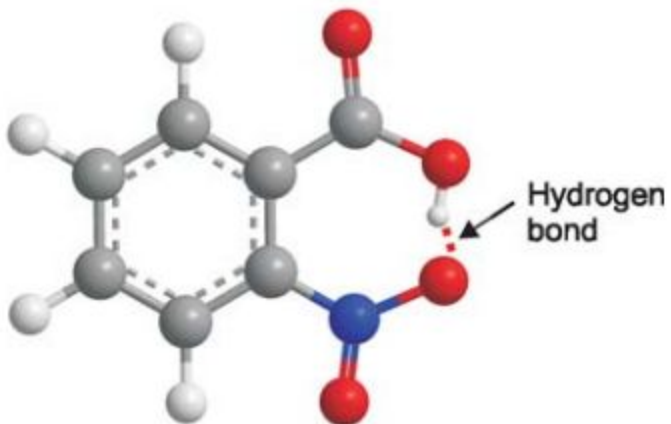
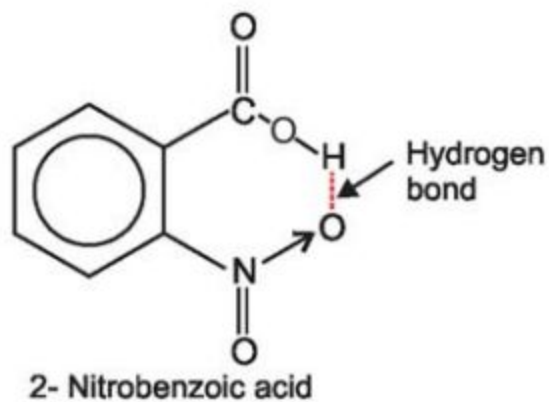
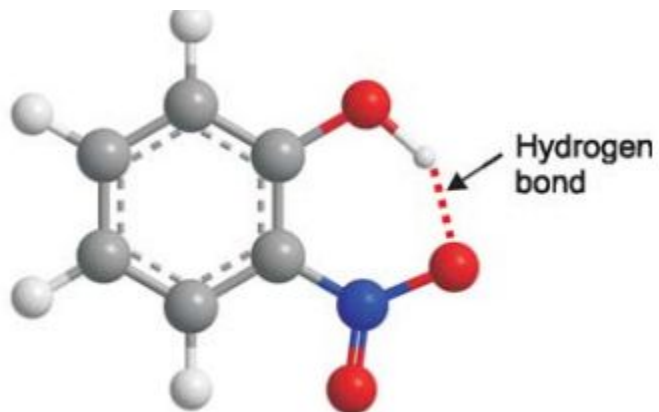
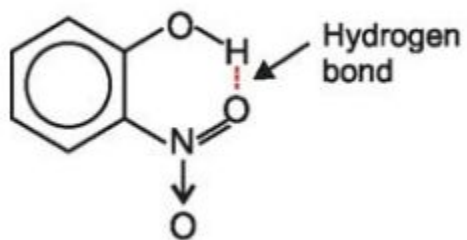
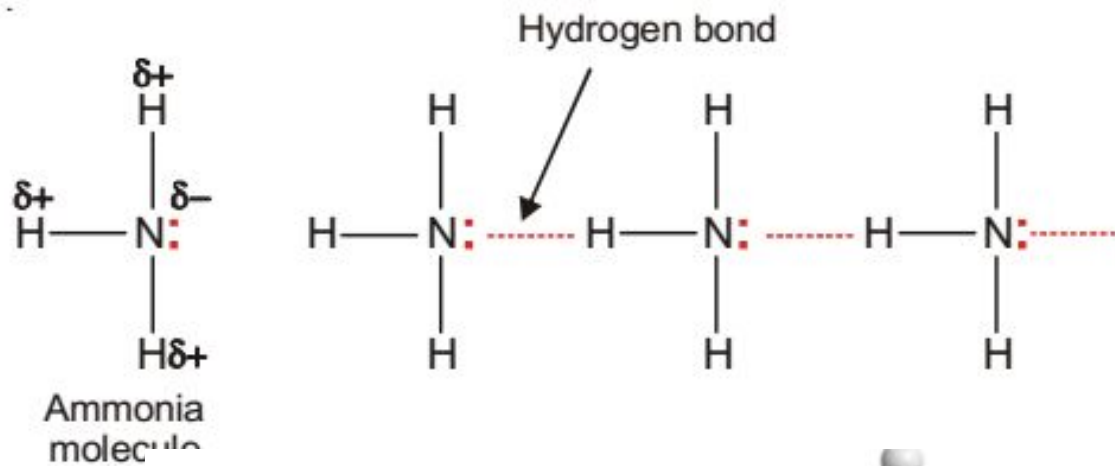
POINTS TO REMEMBER

- (1) Only **O, N and F** which have very high electronegativity and small atomic size, are capable of forming hydrogen bonds.
- (2) Hydrogen bond is longer and much weaker than a normal covalent bond. Hydrogen bond energy is less than 10 kcal/mole, while that of covalent bond is about 120 kcal/mole.
- (3) Hydrogen bonding results in long chains or clusters of a large number of 'associated' molecules like many tiny magnets.
- (4) Like a covalent bond, hydrogen bond has a preferred bonding direction. This is attributed to the fact that hydrogen bonding occurs through p orbitals which contain the lone pair of electrons on X atom. This implies that the atoms X–H...X will be in a straight line.

EXAMPLES OF HYDROGEN-BONDED COMPOUNDS

When hydrogen bonding occurs between different molecules of the same compound as in HF, H₂O and NH₃, it is called **Intermolecular hydrogen bonding**. If the hydrogen bonding takes place within single molecule as in 2-nitrophenol, it is referred to as **Intramolecular hydrogen bonding**. We will consider examples of both types.





CHARACTERISTICS OF HYDROGEN-BONDED COMPOUNDS

(1) Abnormally high boiling and melting points

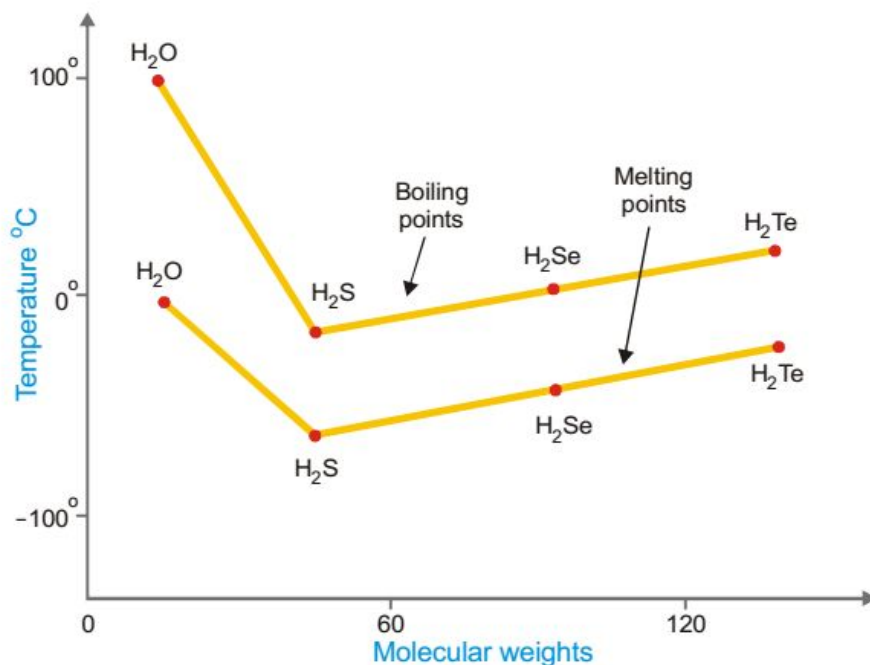
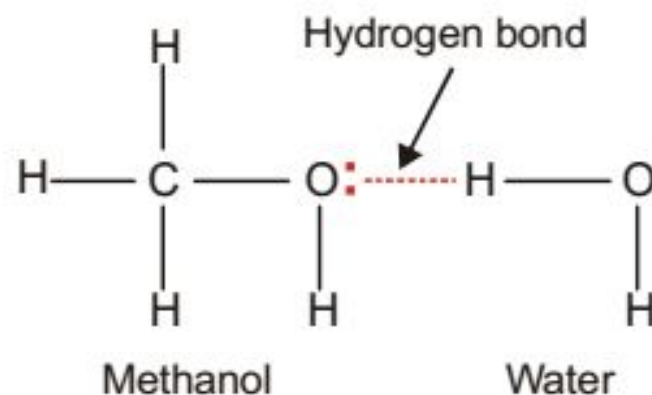
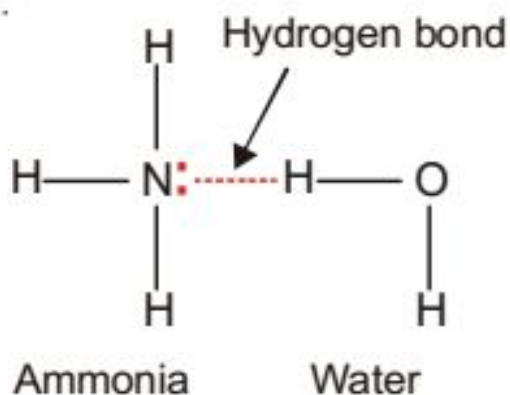


Figure 5.8

Boiling and melting point curves of the hydrides of VIA group showing abrupt increase for water (H₂O) although it has the lowest molecular weight.

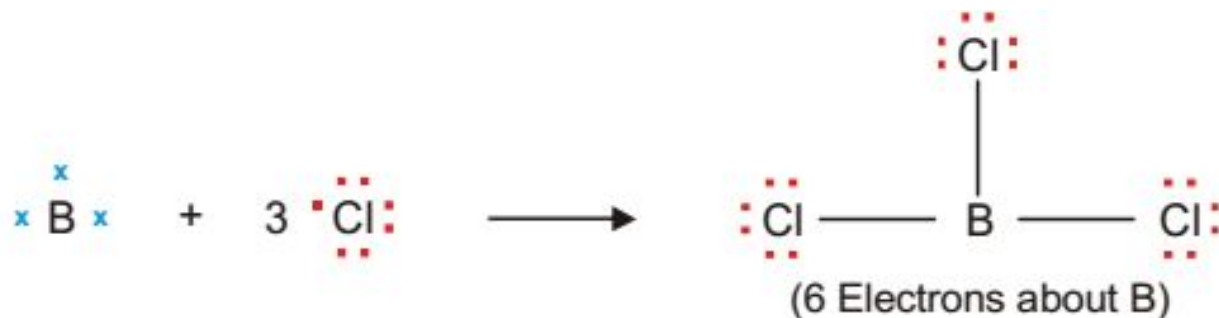
(2) High solubilities of some covalent compounds

The unexpectedly high solubility's of some compounds containing O, N and F, such as NH_3 and CH_3OH in certain hydrogen containing solvents are due to hydrogen bonding. For example, ammonia (NH_3) and methanol (CH_3OH) are highly soluble in water as they form hydrogen bonds.

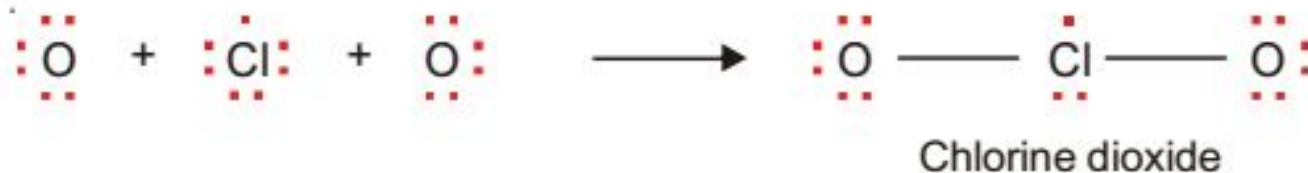


Exceptions To The Octet Rule

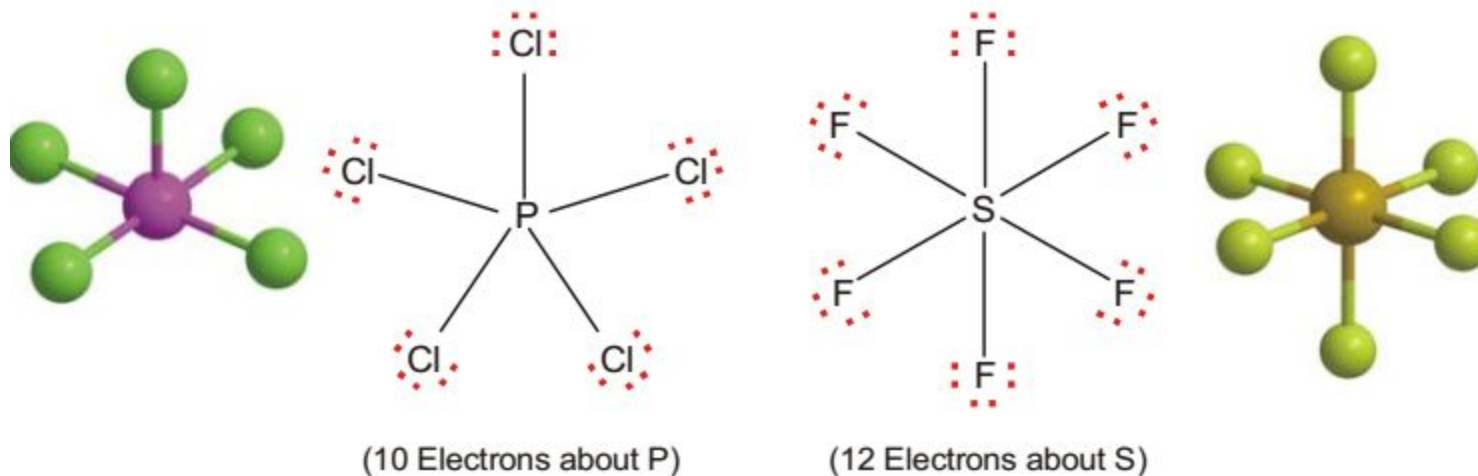
- Four or six electrons around the central atom



- Seven electrons around the central atom



- Ten or more electrons around the central atom



The molecules with more than an octet of electrons are called superoctet structures.

Metallic Bonding

The valence bonds that hold the atoms in a metal crystal together are not ionic, nor are they simply covalent in nature. Ionic bonding is obviously impossible here since all the atoms would tend to give electrons but none are willing to accept them. Ordinary covalent bonding is also ruled out as, for example, sodium atom with only one outer-shell electron could not be expected to form covalent bonds with 8 nearest neighboring atoms in its crystal.

The peculiar type of bonding which holds the atoms together in metal crystal is called the Metallic Bonding.

THE ELECTRON SEA MODEL

Metal atoms are characterized by :

- (1) Low ionization energies which imply that the valence electrons in metal atoms can easily be separated.
- (2) A number of vacant electron orbitals in their outermost shell. For example, the magnesium atom with the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^0$ has three vacant 3p orbitals in its outer electron shell.

There is considerable overlapping of vacant orbitals on one atom with similar orbitals of adjacent atoms, throughout the metal crystal. Thus it is possible for an electron to be delocalized and move freely in the vacant molecular orbital encompassing the entire metal crystal. The delocalized electrons no longer belong to individual metal atoms but rather to the crystal as a whole. As a result of the delocalization of valence electrons, the positive metal ions that are produced, remain fixed in the crystal lattice while the delocalized electrons are free to move about in the vacant space in between. The metal is thus pictured as a network or lattice of positive ions of the metal immersed in a 'sea of electrons' or 'gas of electrons'. This relatively simple model of metallic bonding is referred to as the Electron Sea model or the Electron Gas model (Fig. 5.10.)

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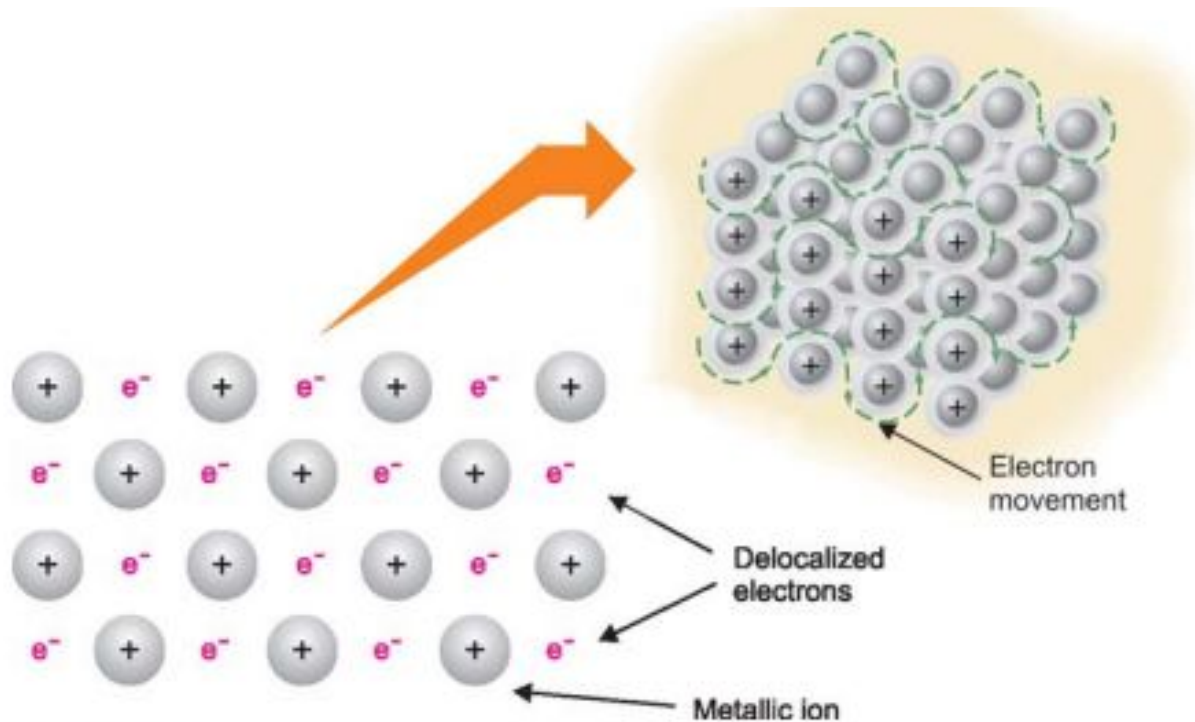


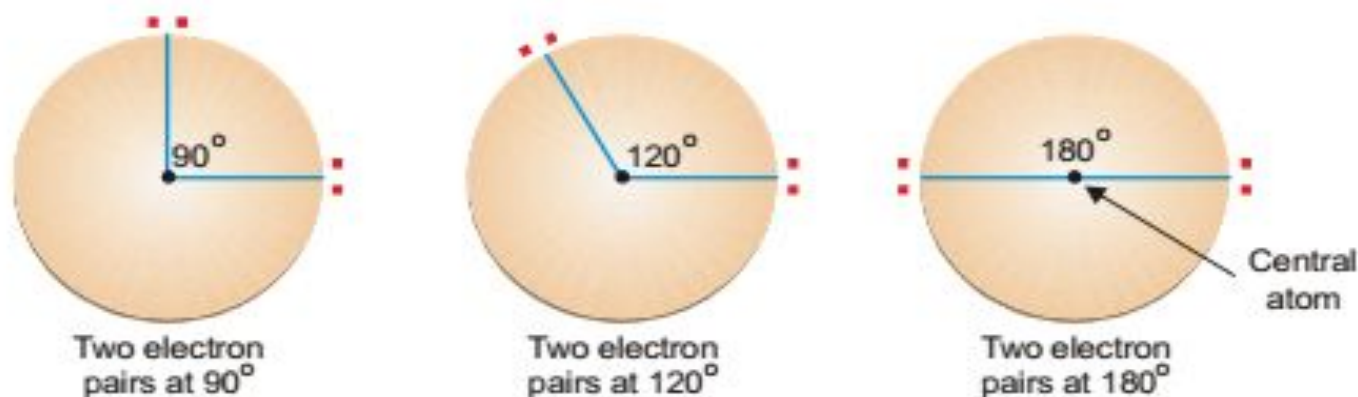
Figure 5.10
The Electron Sea model of metallic bonding.

The electron sea model of metallic bonding explains fairly well the most characteristic physical properties of metals.

- (1) Luster or Reflectivity.
- (2) Electric Conductivity.
- (3) Heat Conductivity.
- (4) Ductility and Malleability.
- (5) Electron Emission.

VSEPR THEORY

The Lewis structure of a molecule tells us the number of pairs of electrons in the valence shell of the central atom. These electron pairs are subject to electrostatic attractions between them. **On this basis, R.G.Gillespie (1970) proposed a theory called the Valence-Shell Electron Pair Repulsion or VSEPR (pronounced as 'Vesper') theory.** It states that : The electron pairs (both lone pairs and shared pairs, surrounding the central atom will be arranged in space as far apart as possible to minimize the electrostatic repulsion between them.



■ **Figure 5.14**

Arrangement of two electron pairs on circle at 90° , 120° and at 180° . Placement of electron pairs at 180° puts them the farthest apart, thereby minimising the electrostatic repulsion.

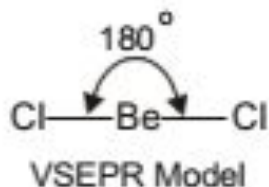
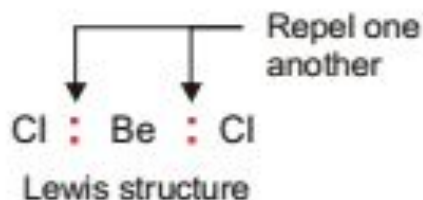
VSEPR theory is simple but remarkably powerful model for predicting molecular geometries and bond angles. While working out the shapes of molecules from this theory, it must be remembered :

- (1) Multiple bonds behave as a single electron-pair bond for the purpose of VSEPR. They represent a single group of electrons.
- 2) Order of repulsions between lone pair and lone pair (lp-lp), lone pair and bonding pair (lp-bp), and bonding pair and bonding pair (bp-bp) is $lp - lp > lp - bp > bp - bp$

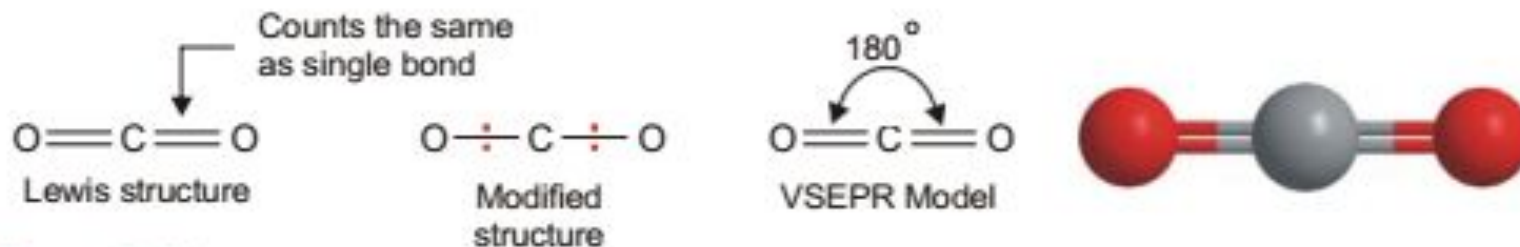
When a molecule has lone pairs of electrons, the bonding electron pairs are pushed closer and thus the bond angle is decreased.

(1) Linear Molecules

(a) Beryllium chloride, BeCl_2 . It has the Lewis structure



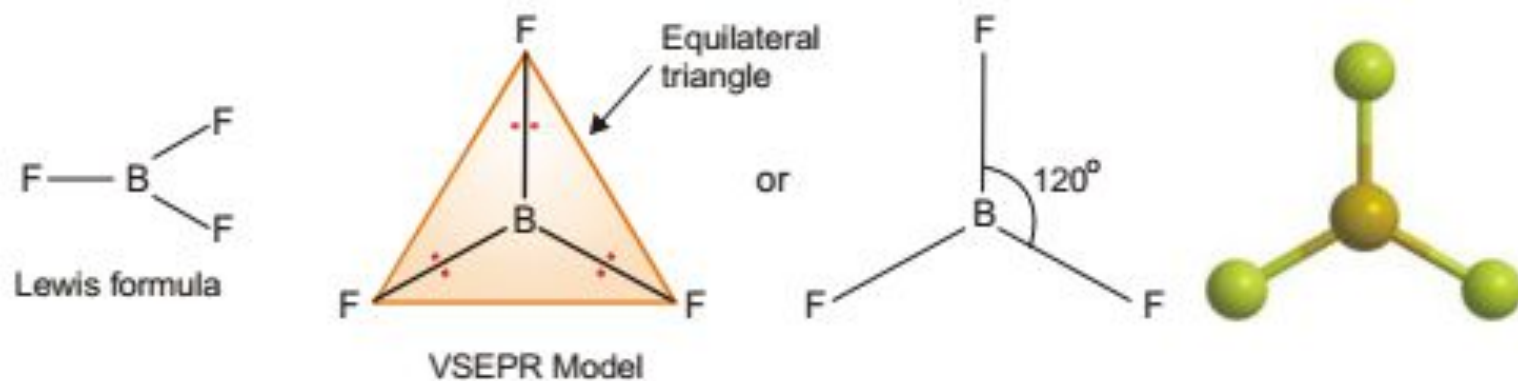
(b) Carbon dioxide, CO_2 . It has the structure



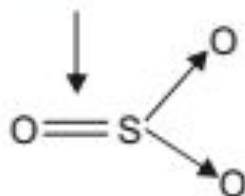
■ **Figure 5.16**
Geometry of CO_2 molecule.

(2) Trigonal Planar Molecules

(a) Boron trifluoride, BF_3 . Its Lewis structure shows that the central atom B has three bonding electron pairs and no unshared electrons. VSEPR theory says that the three bonding electron pairs will be as far apart as possible. This can be so if these electron pairs are directed to the corners of an equilateral triangle. Thus VSEPR model of BF_3 molecule has three F atoms at the corners of the triangle with B atom at its center. All the four atoms (three F and one B) lie in the same plane. Therefore, the shape of such a molecule is called trigonal planar. The bond angle is 120° .

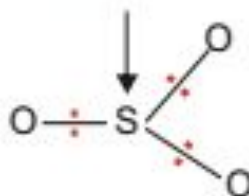


Counted as
single electron pair

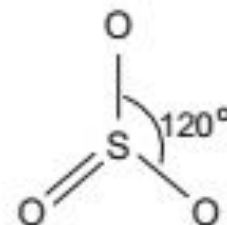


Lewis structure

Surrounded by
3 electron pairs



Modified
structure



VSEPR Model
(trigonal, planar)

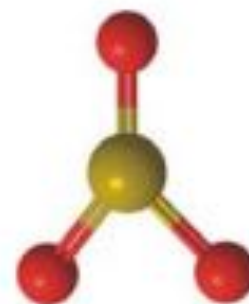
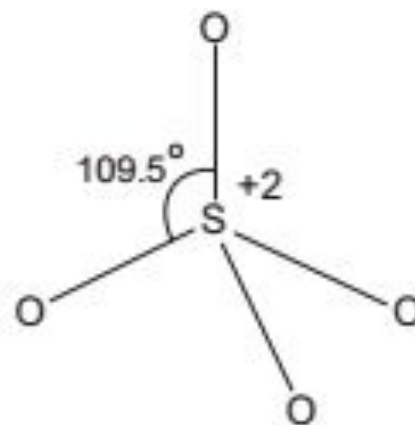
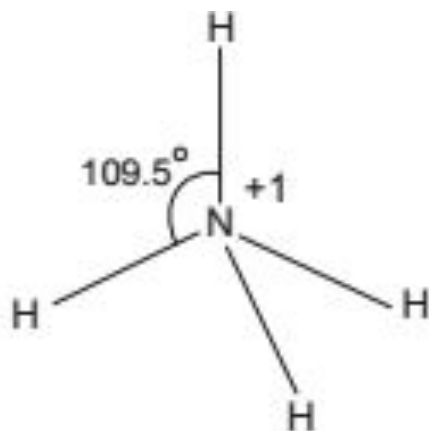
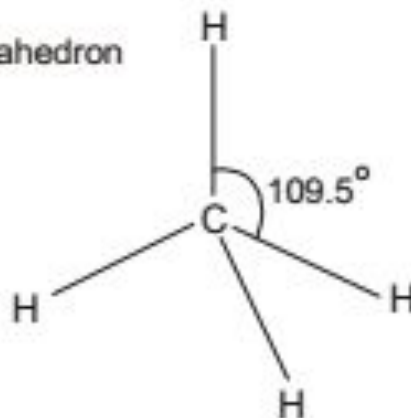
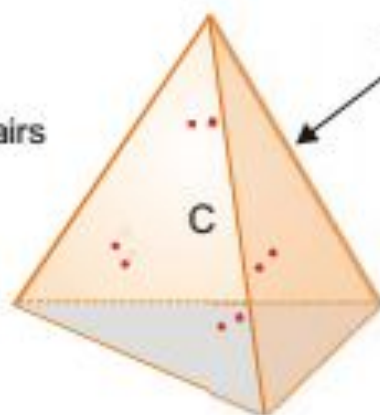
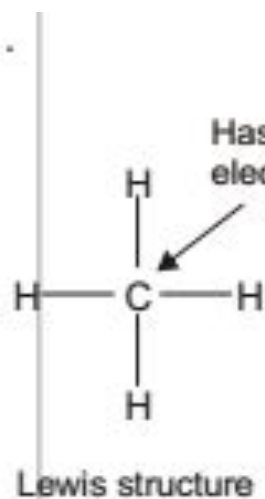
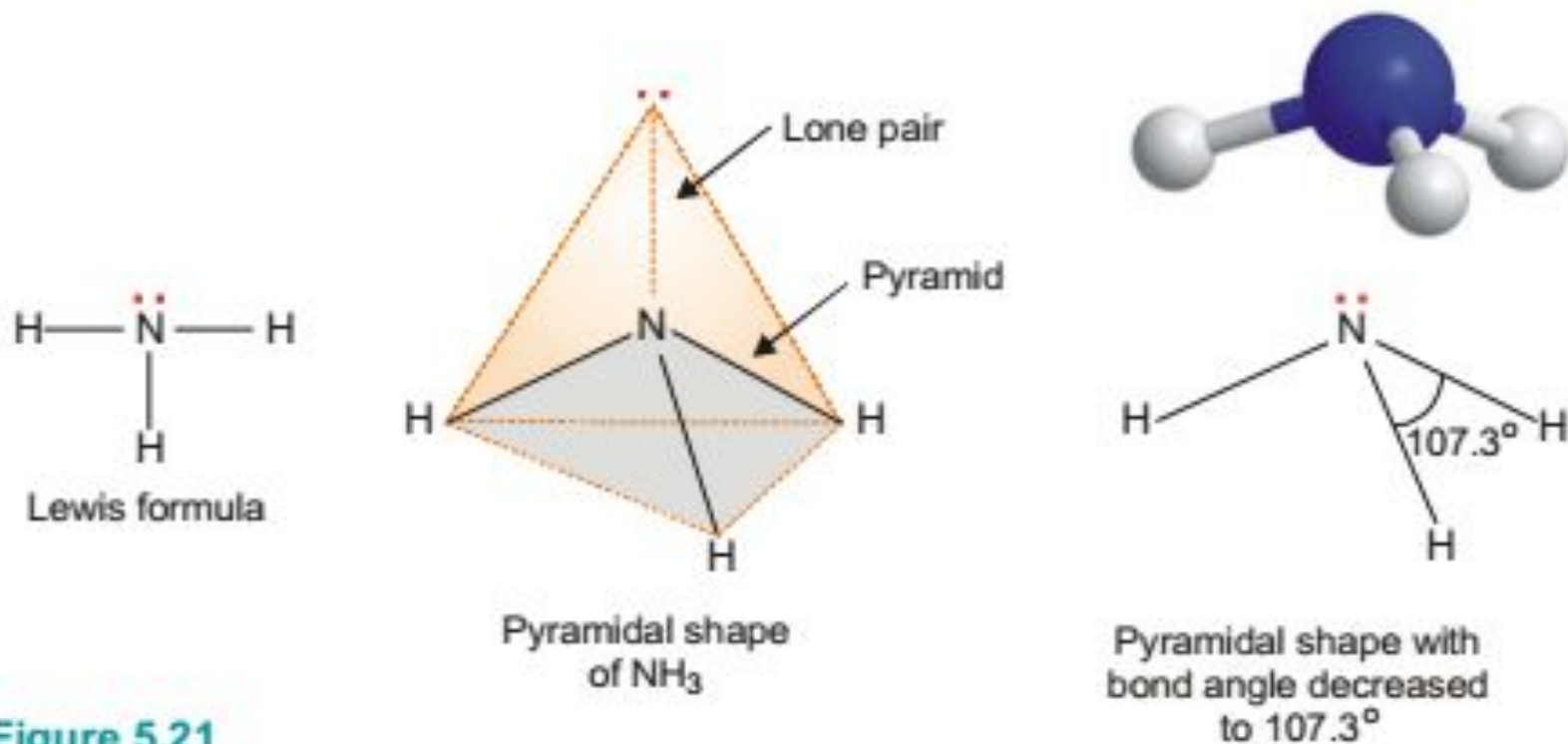


Figure 5.18
 Geometry of SO_3 molecule.

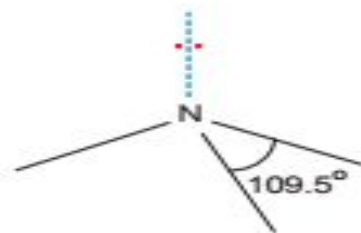
(3) Tetrahedral



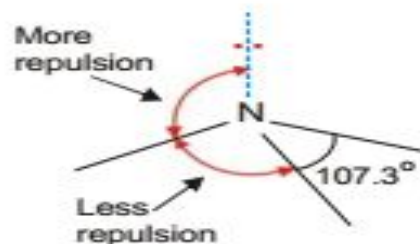
According to VSEPR theory, a lone pair exerts greater repulsion on the bonding electron pairs than the bonding pairs do on each other. As a result, the bonds of NH_3 molecule are pushed slightly closer. This explains why the observed bond angle $\text{H}-\text{N}-\text{H}$ is found to be 107.3° instead of 109.5° predicted from tetrahedral geometry.



■ **Figure 5.21**
Geometry of NH_3 molecule.



Tetrahedral geometry predicts H—N—H angle of 109.5°

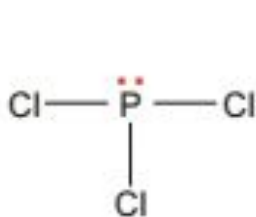


The bond angle decreases because of greater repulsion of lone electron pair

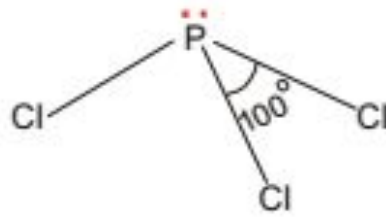
■ **Figure 5.22**

Why the angle H—N—H in NH_3 molecule is 107.3° while the tetrahedral angle is 109.5° ?

(b) Phosphorus trichloride, PCl_3 . The structural formula indicates that the central phosphorus atom has three bonding electron pairs and one lone electron pair. Thus, like NH_3 it has pyramidal shape and the observed bond angle Cl—P—Cl is 100° .



Lewis formula

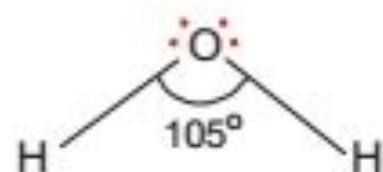
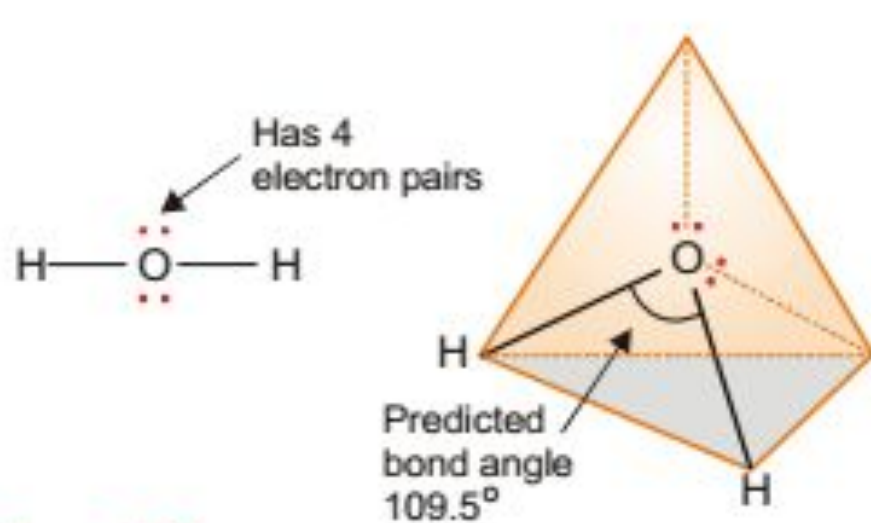


Pyramidal PCl_3 molecule



■ **Figure 5.23**

Geometry of PCl_3 molecule.



Bent molecule of H_2O with observed bond angle 105°

Figure 5.24
Geometry of H_2O molecule.

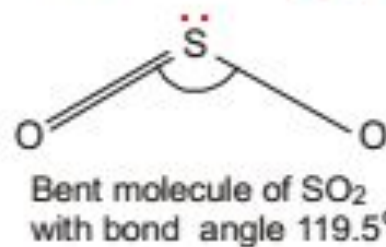
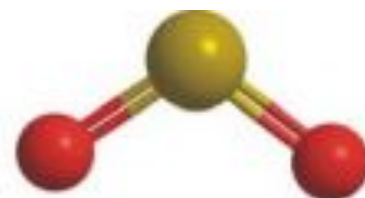
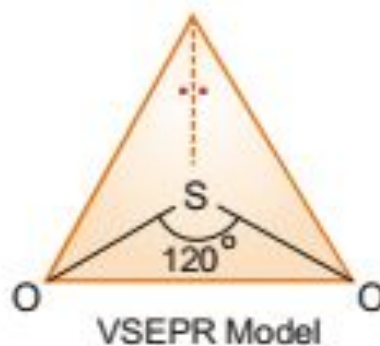
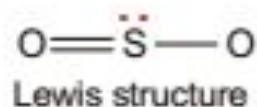
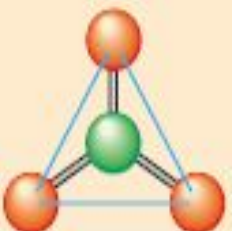
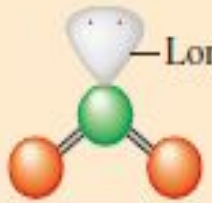


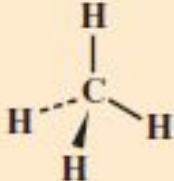
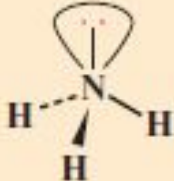
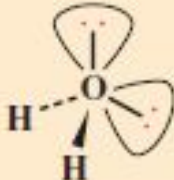
Figure 5.25
Geometry of SO_2 molecule.

Summery of geometry

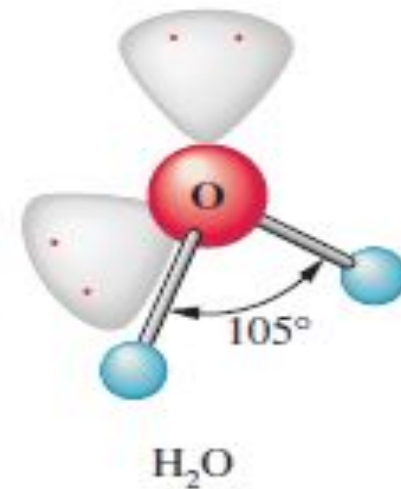
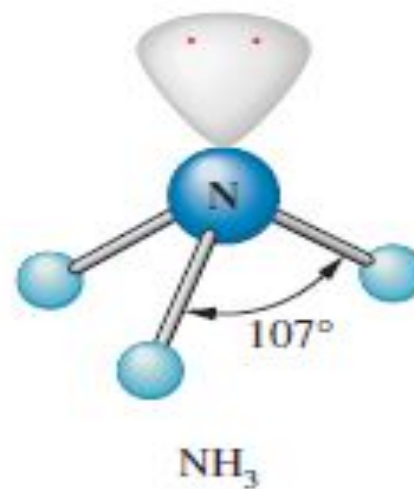
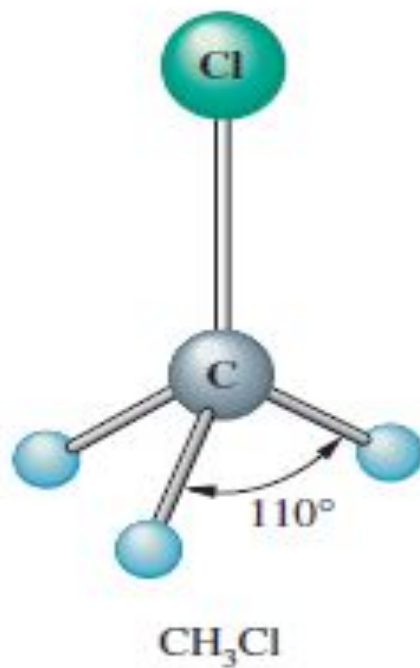
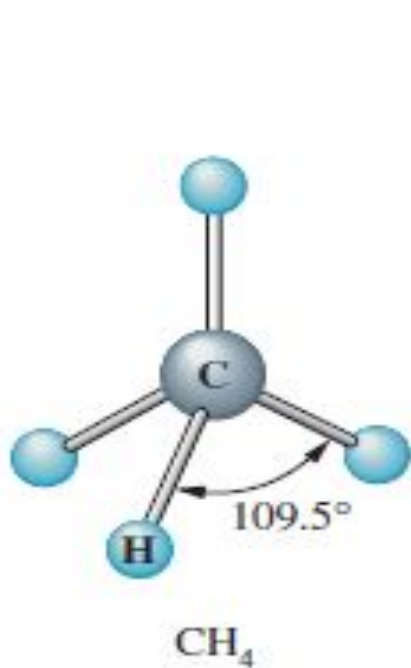
Central Atom with Two and Three Valence-Shell Electron Pairs

ELECTRON PAIRS			ARRANGEMENT OF PAIRS	MOLECULAR GEOMETRY	EXAMPLE
Total	Bonding	Lone			
2	2	0	Linear	Linear AX_2	 BeF_2 $F-Be-F$
3	$\left\{ \begin{array}{c} 3 \\ 2 \end{array} \right.$	$\left\{ \begin{array}{c} 0 \\ 1 \end{array} \right.$	Trigonal planar	Trigonal planar AX_3	 BF_3 
				Bent (or angular) AX_2	 Lone pair SO_2 

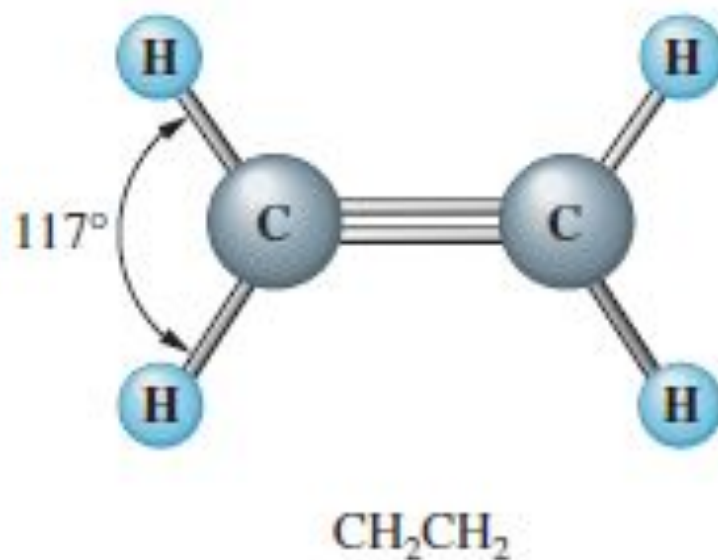
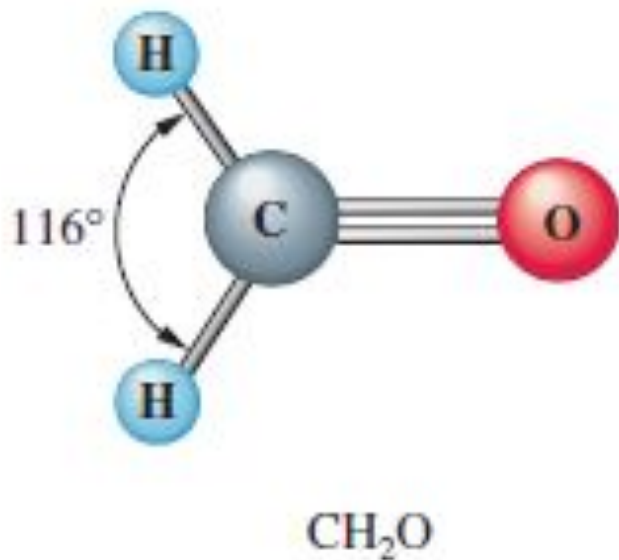
Central Atom with Four Valence-Shell Electron Pairs

ELECTRON PAIRS			ARRANGEMENT OF PAIRS	MOLECULAR GEOMETRY	EXAMPLE
Total	Bonding	Lone			
4	4	0	Tetrahedral	Tetrahedral AX_4	CH_4 
	3	1		Trigonal pyramidal AX_3	NH_3 
	2	2		Bent (or angular) AX_2	H_2O 

A lone pair tends to require more space than a corresponding bonding pair.



Multiple bonds require more space than single bonds because of the greater number of electrons.



Central Atom with Five Valence-Shell Electron Pairs

ELECTRON PAIRS
Total Bonding Lone

**ARRANGEMENT
OF PAIRS**

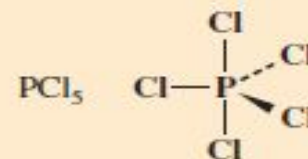
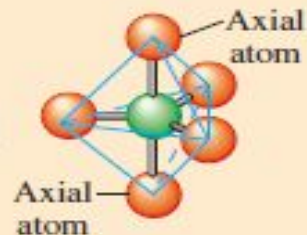
**MOLECULAR
GEOMETRY**

EXAMPLE

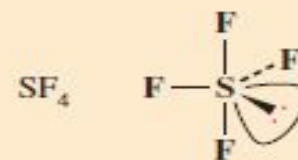
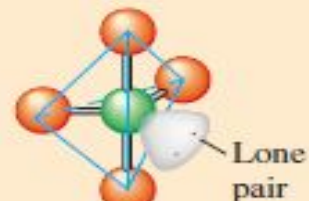
5	5	0
	4	1
	3	2
	2	3

Trigonal
bipyramidal

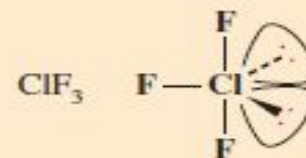
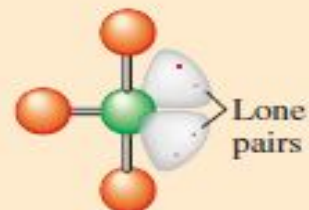
Trigonal
bipyramidal
 AX_5



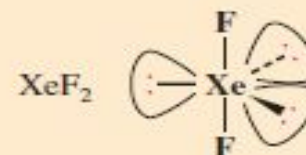
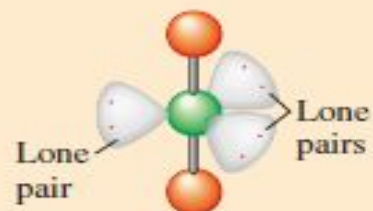
Seesaw
(or distorted
tetrahedron)
 AX_4



T-shaped
 AX_3



Linear
 AX_2



Central Atom with Six Valence-Shell Electron Pairs

ELECTRON PAIRS			ARRANGEMENT OF PAIRS	MOLECULAR GEOMETRY	EXAMPLE
Total	Bonding	Lone			
6	6	0	Octahedral	Octahedral AX_6	SF_6 
	5	1		Square pyramidal AX_5	IF_5 
	4	2		Square planar AX_4	XeF_4 

Valence Bond Theory

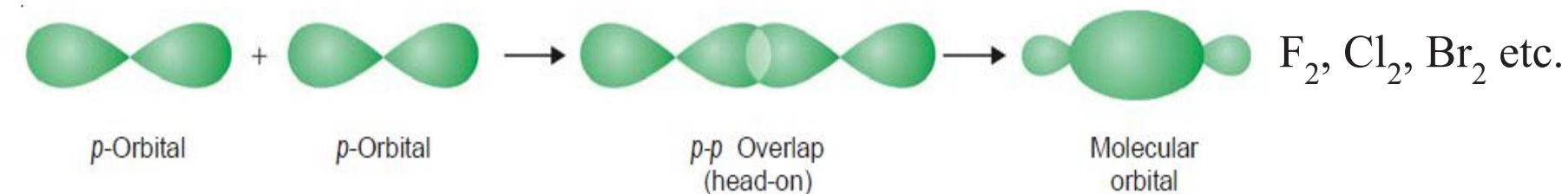
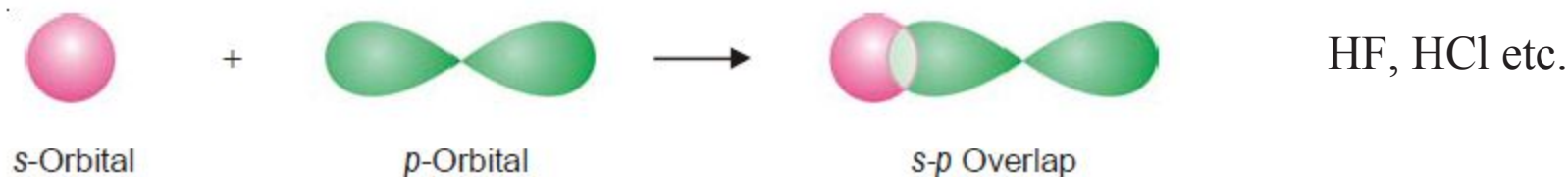
The VSEPR model is usually a satisfactory method for predicting molecular geometries. To understand bonding and electronic structure, however, you must look to quantum mechanics. We will consider two theories stemming from quantum mechanics: valence bond theory and molecular orbital theory

1. An orbital on one atom comes to occupy a portion of the same region of space as an orbital on the other atom. The two orbitals are said to overlap.
2. The total number of electrons in both orbitals is not more than two.

Types of overlapping and Nature of Covalent Bond

A covalent bond is of two types depending on the type of overlapping between the two atoms

(1) Sigma (σ) bond; (2) Pi (π) Bond



1) Sigma (σ) Bond

When there is end to end overlapping of atomic orbitals along the internuclear axis, the bond resulted is called sigma (σ) bond. This type of overlapping between the atomic orbitals is also called “head-on” overlapping or “axial” overlapping.

(a) s – s overlapping

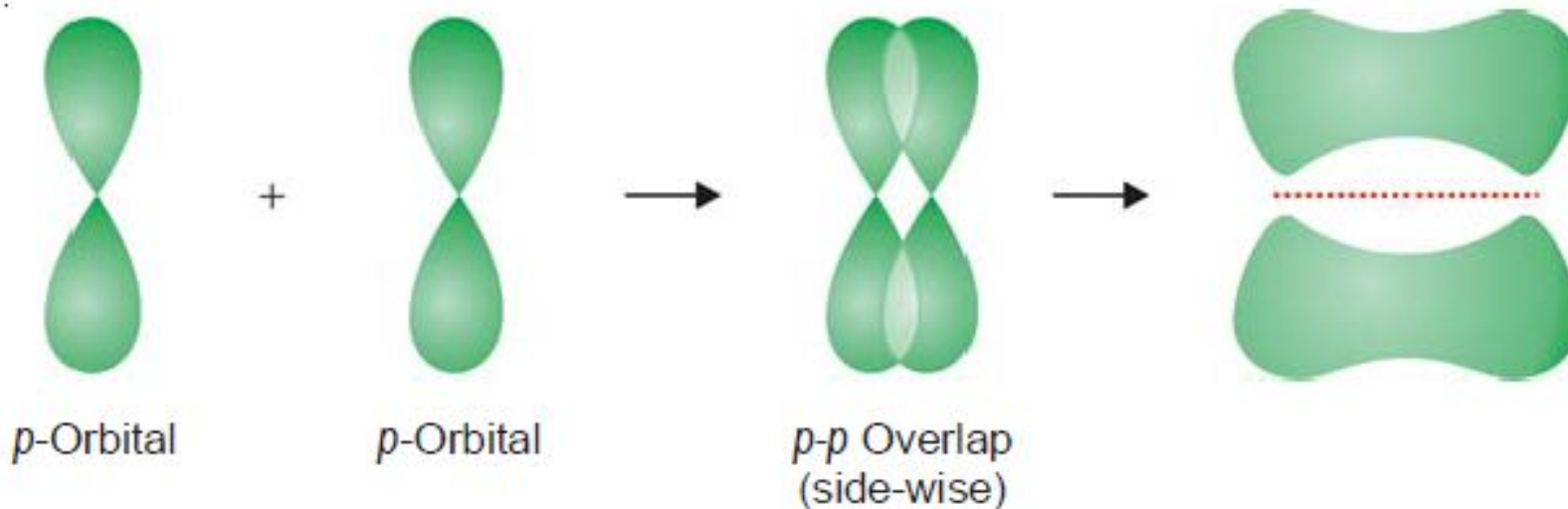
Here s-orbital of one atom overlaps with the s-orbital of other atom.

(b) s – p overlapping

In this type of overlap s-orbital of one atom overlaps with the half filled p-orbital of the other atom.

(2) Pi (π) Bond

This type of covalent bond is formed by the sidewise overlap of the half filled atomic orbitals. It is also called lateral or sidewise overlap. This type of overlapping takes place perpendicular to the internuclear axis as shown below



Sigma (σ) bond

1. It is formed by *end to end* overlapping of half filled atomic orbitals.
2. Overlapping takes place along internuclear axis.
3. The extent of overlapping is large and bond formed is *stronger*.
4. The molecular orbital formed as a result of overlapping is symmetrical about the internuclear axis.
5. There is free rotation about σ bond and no geometrical isomers are possible.
6. The bond can be present alone.
7. *s* and *p* orbitals can participate in the formation of σ bond.

Pi (π) Bond

1. It is formed by the sidewise overlapping of half filled *p*-orbitals only.
2. Overlapping takes place perpendicular to internuclear axis.
3. The extent of overlapping is small and bond formed is *weaker*.
4. The molecular orbital formed as a result of overlapping consists of two lobes above and below the internuclear axis.
5. There is no free rotation about π bond and geometrical isomers are possible.
6. The bond is always formed in addition to sigma (σ) bond.
7. Only *p*-orbitals participate in the formation of π bond.

Hybrid Orbitals

From what has been said, you might expect **the number of bonds formed by a given atom to equal the number of unpaired electrons** in its valence shell. Chlorine, whose orbital diagram is



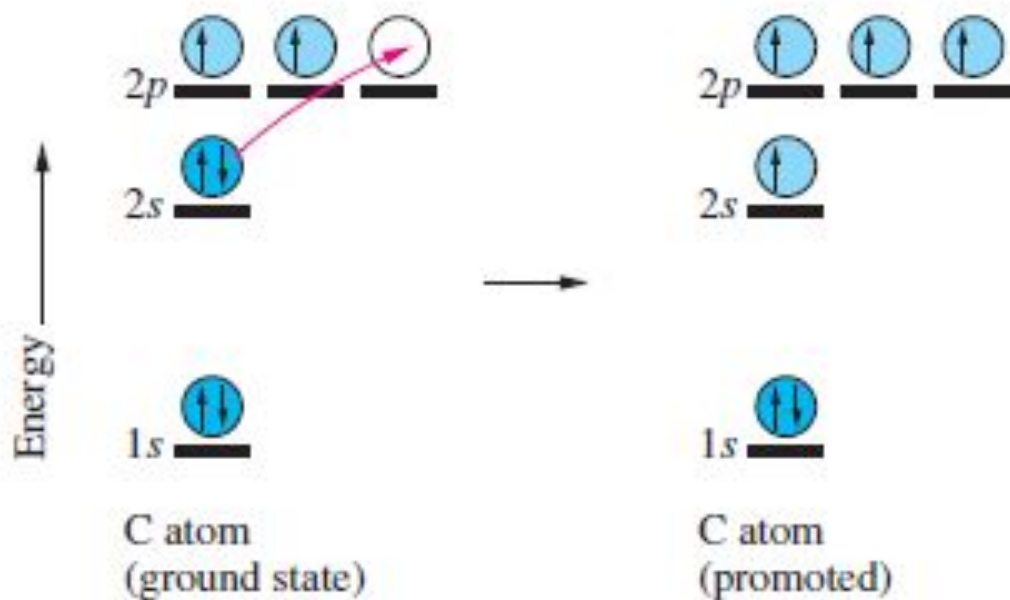
has one unpaired electron and forms one bond. Oxygen, whose orbital diagram is



has two unpaired electrons and forms two bonds, as in H_2O . However, consider the carbon atom, whose orbital diagram is



You might expect this atom to bond to two hydrogen atoms to form the CH_2 molecule. Although this molecule is known to be present momentarily during some reactions, it is very reactive and cannot be isolated. But methane, CH_4 , in which the carbon atom bonds to four hydrogen atoms, is well known. In fact, a carbon atom usually forms four bonds.

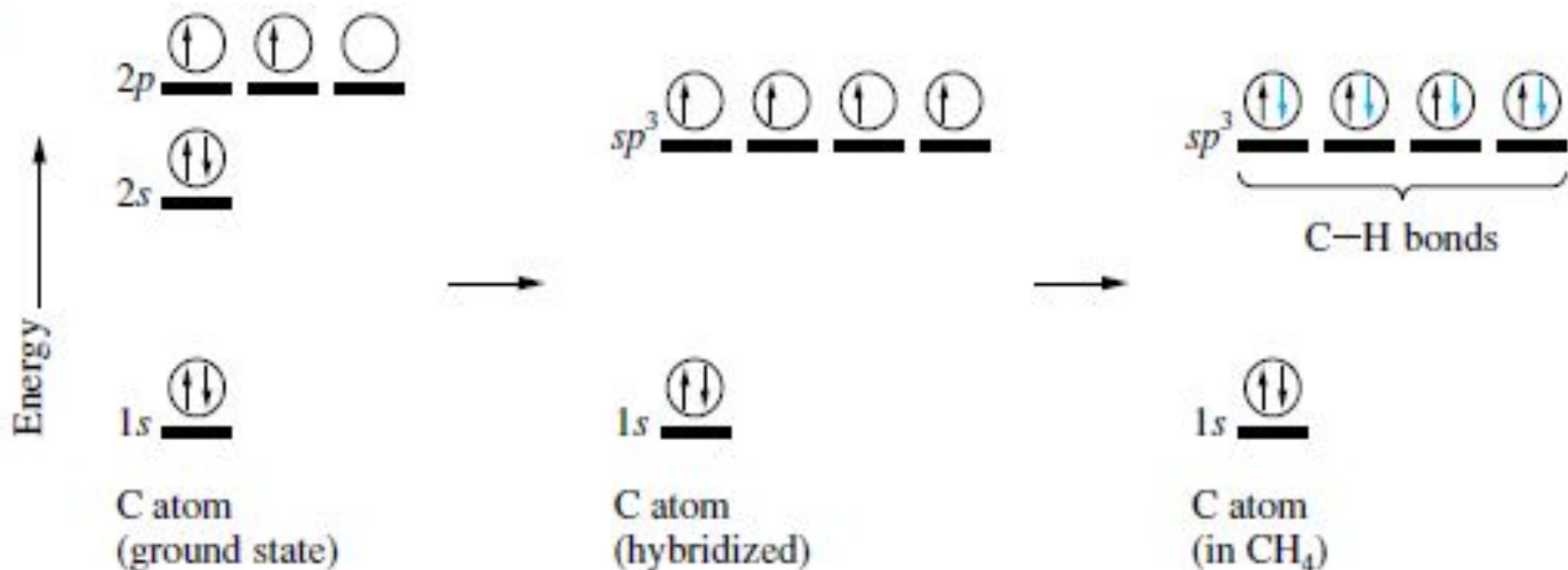


Nuclear magnetic resonance and infrared spectroscopy both show that CH_4 has four equivalent C+H bonds.

Hybrid orbitals are orbitals used to describe bonding that are obtained by taking combinations of atomic orbitals of the isolated atoms.

Hybrid orbitals are orbitals used to describe bonding that are obtained by taking combinations of atomic orbitals of the isolated atoms.

In this case, a set of hybrid orbitals is constructed from one s orbital and three p orbitals, so they are called sp^3 hybrid orbitals. Calculations from theory show that each sp^3 hybrid orbital has a large lobe pointing in one direction and a small lobe pointing in the opposite direction. The four sp^3 hybrid orbitals point in tetrahedral directions



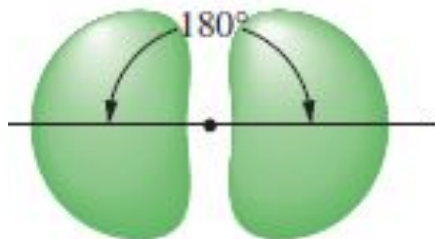
The number of hybrid orbitals formed always equals the number of atomic orbitals used.

Thus hybridization may precisely be defined as the phenomenon of mixing up (or merging) of orbitals of an atom of nearly equal energy, giving rise to entirely new orbitals equal in number to the mixing orbitals and having same energy contents and identical shapes.

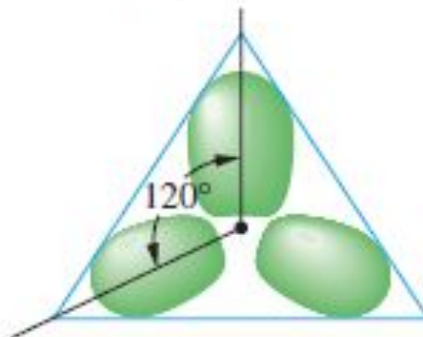
Kinds of Hybrid Orbitals

Hybrid Orbitals	Geometric Arrangement	Number of Orbitals	Example
sp	Linear	2	Be in BeF_2
sp^2	Trigonal planar	3	B in BF_3
sp^3	Tetrahedral	4	C in CH_4
sp^3d	Trigonal bipyramidal	5	P in PCl_5
sp^3d^2	Octahedral	6	S in SF_6

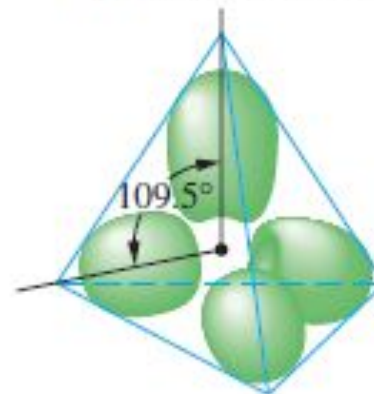
Linear arrangement:
 sp hybrid orbitals



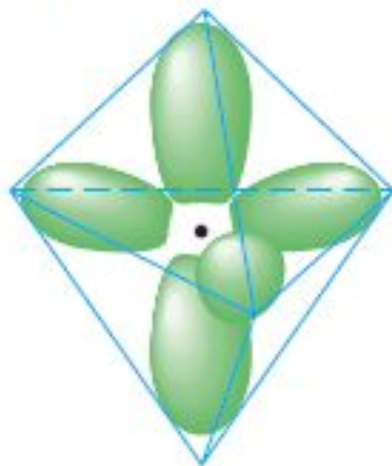
Trigonal planar arrangement:
 sp^2 hybrid orbitals



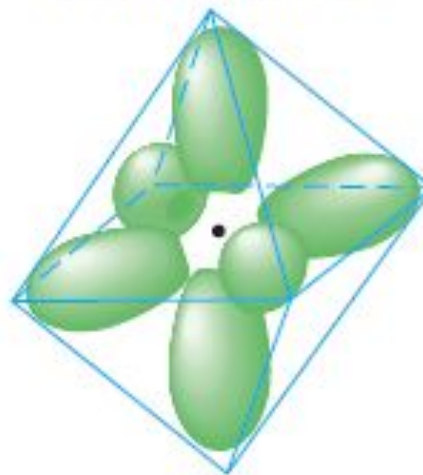
Tetrahedral arrangement:
 sp^3 hybrid orbitals



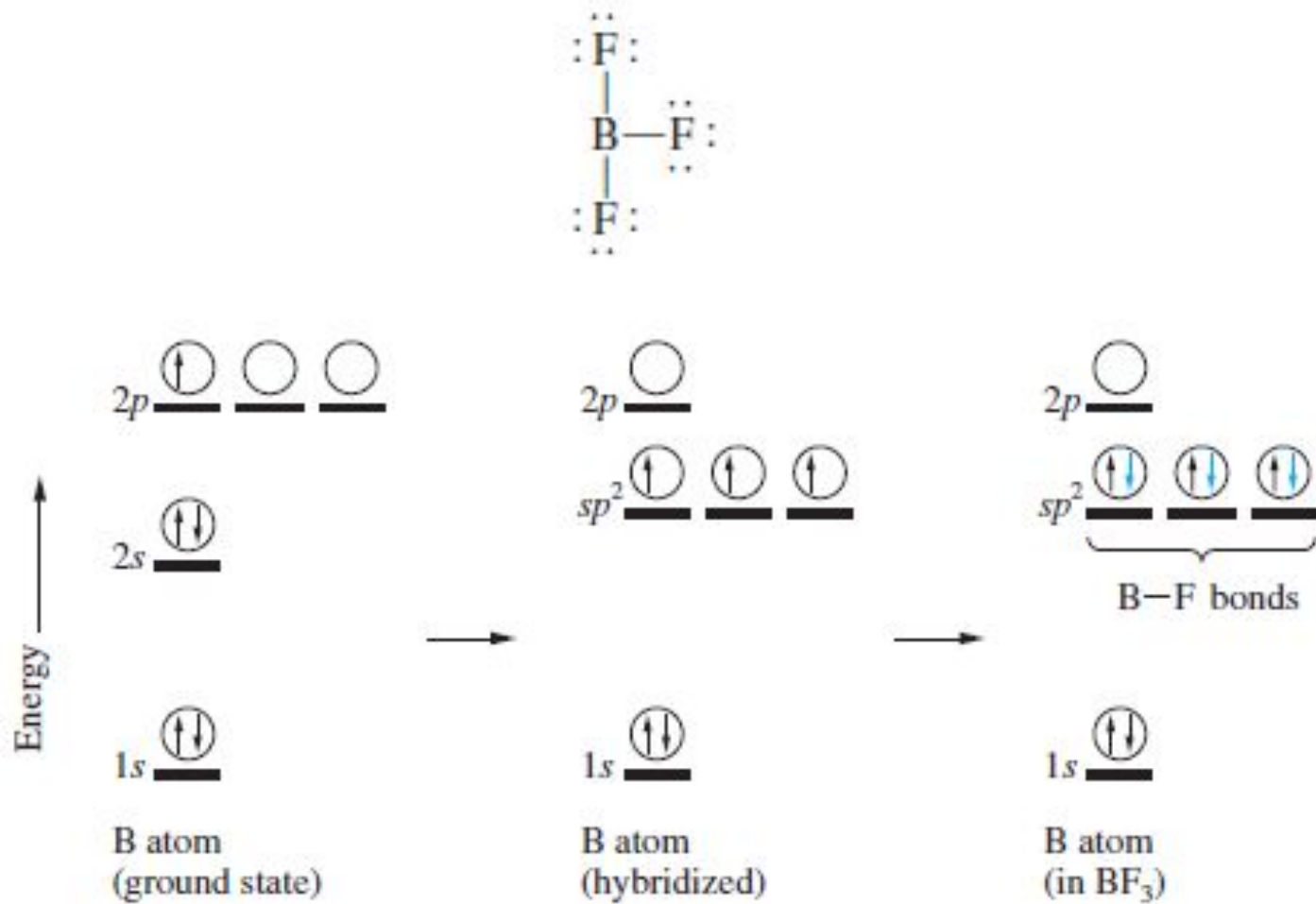
Trigonal bipyramidal arrangement:
 sp^3d hybrid orbitals



Octahedral arrangement:
 sp^3d^2 hybrid orbitals

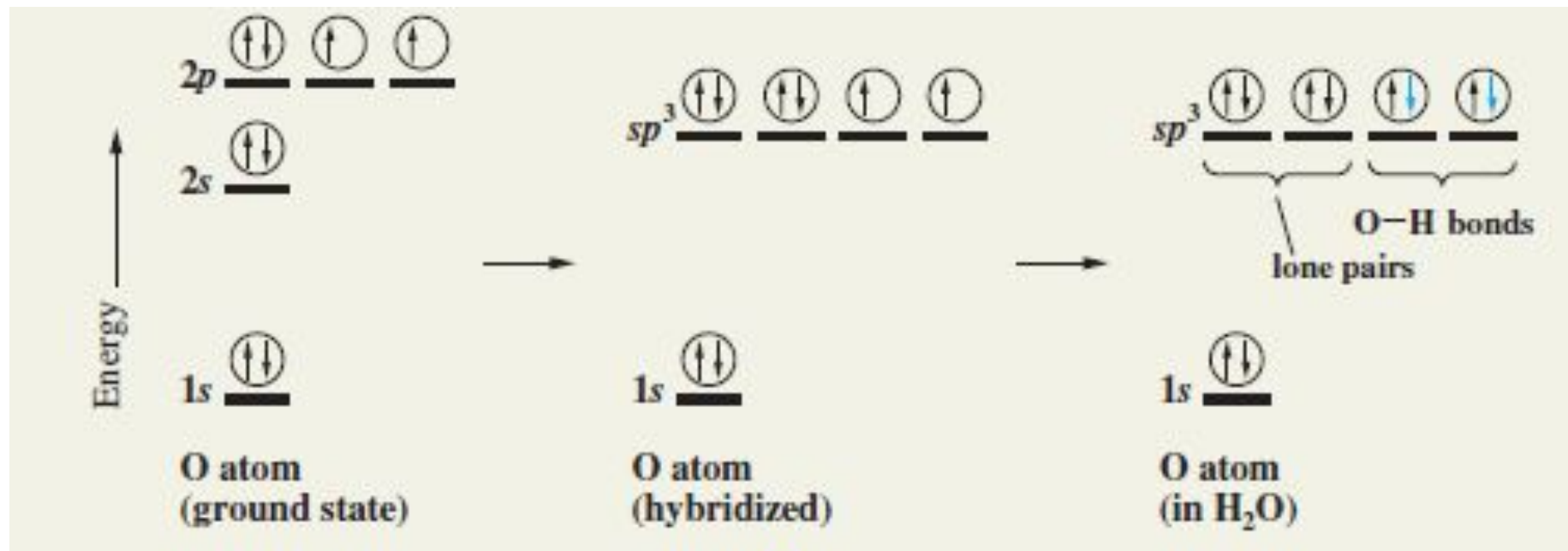
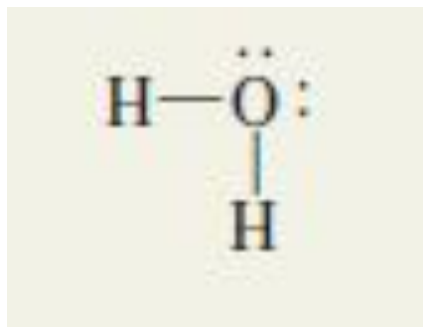
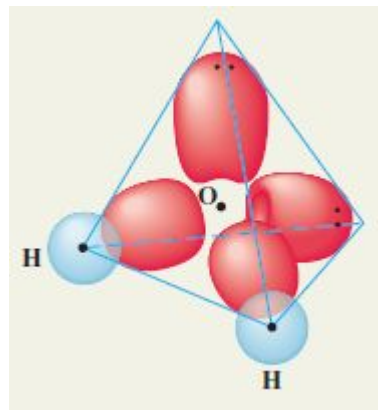


Bonding In BF_3



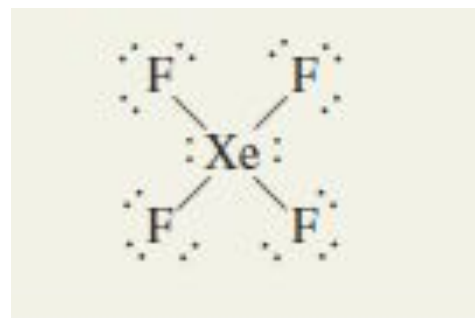
Describe the bonding in H_2O according to valence bond theory. Assume that the molecular geometry is the same as given by the VSEPR model.

The Lewis formula for H_2O is

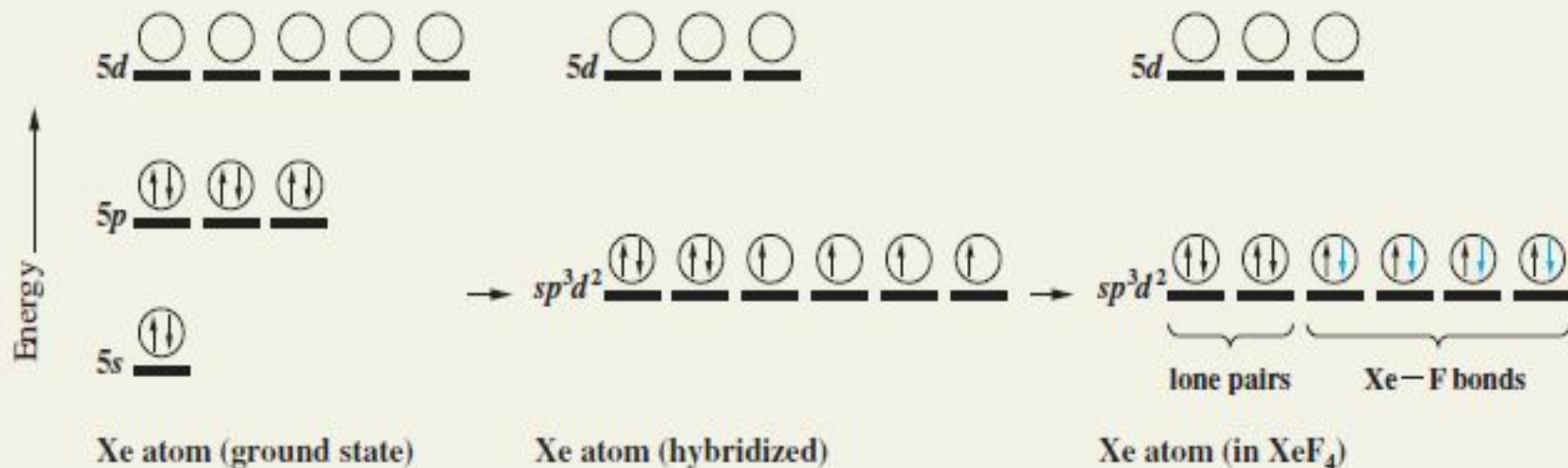


Describe the bonding in XeF_4 using hybrid orbitals.

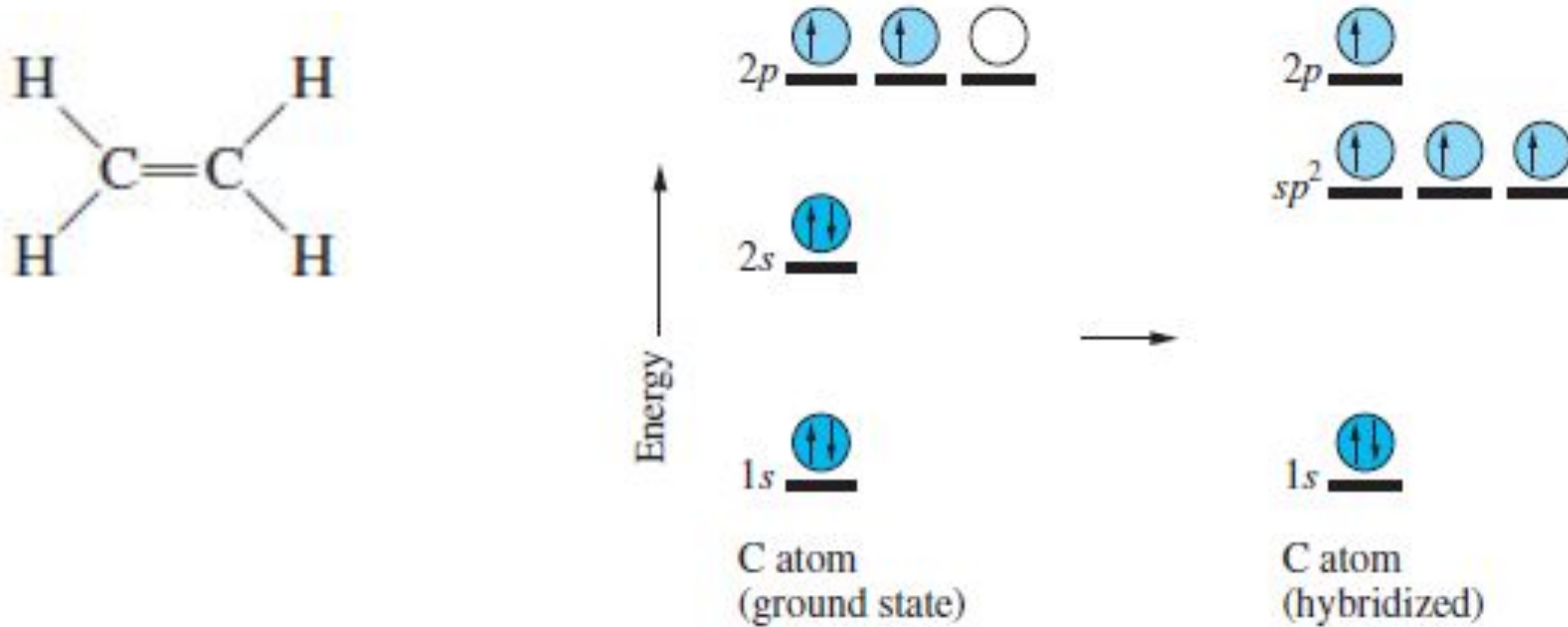
The Lewis formula of XeF_4 is

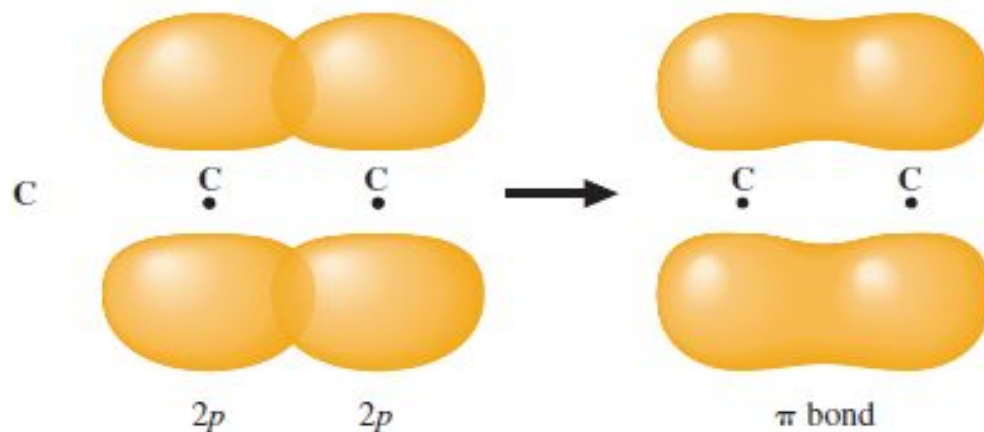
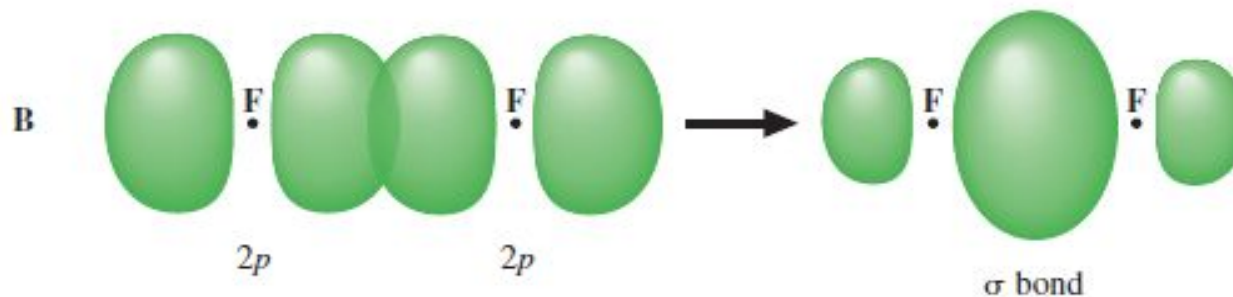
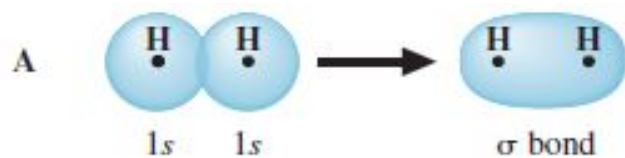


a singly occupied fluorine $2p$ orbital. You can summarize this as follows:

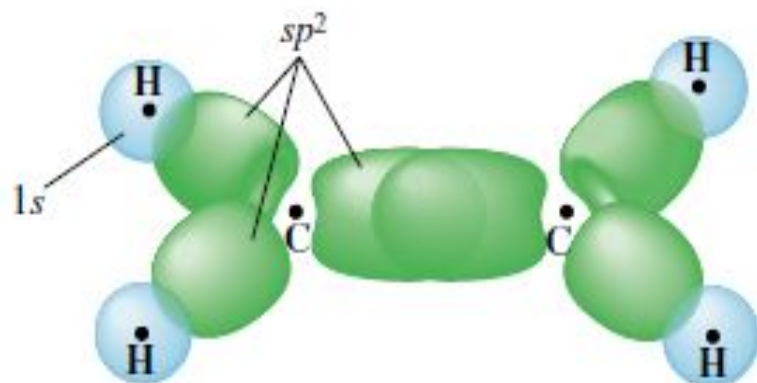


Description of Multiple Bonding

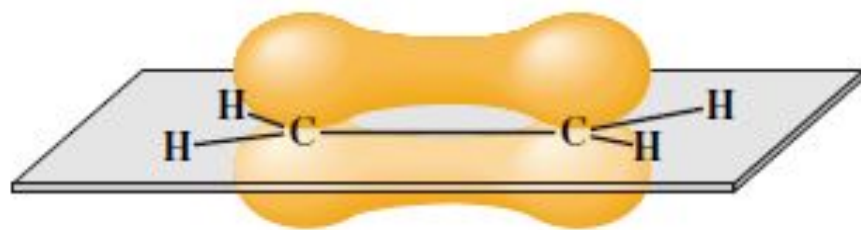
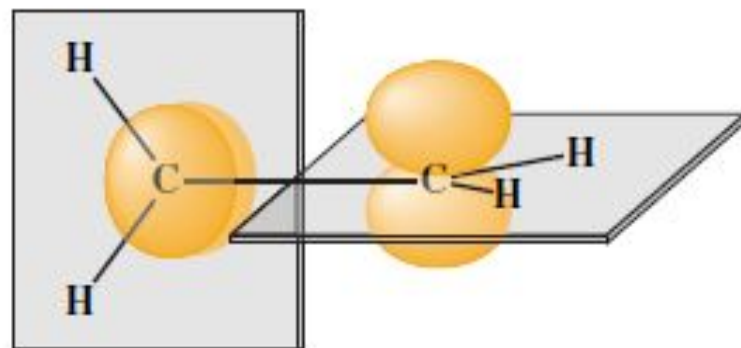




A) The formation of a σ bond by the overlap of two s orbitals. (B) A σ bond can also be formed by the overlap of two p orbitals along their axes. (C) When two p orbitals overlap sideways, a π bond is formed.

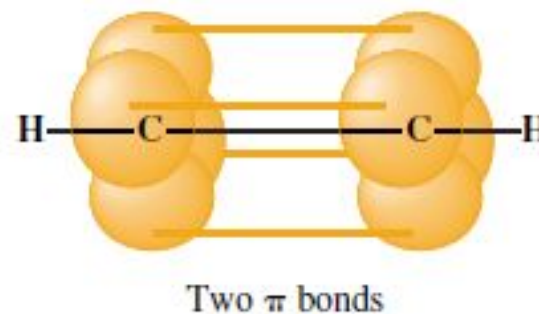
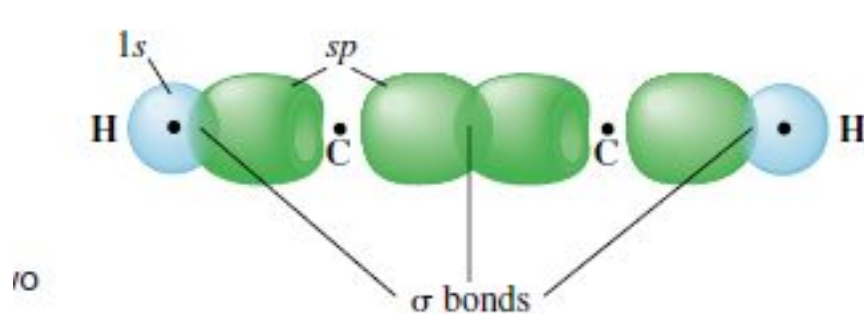
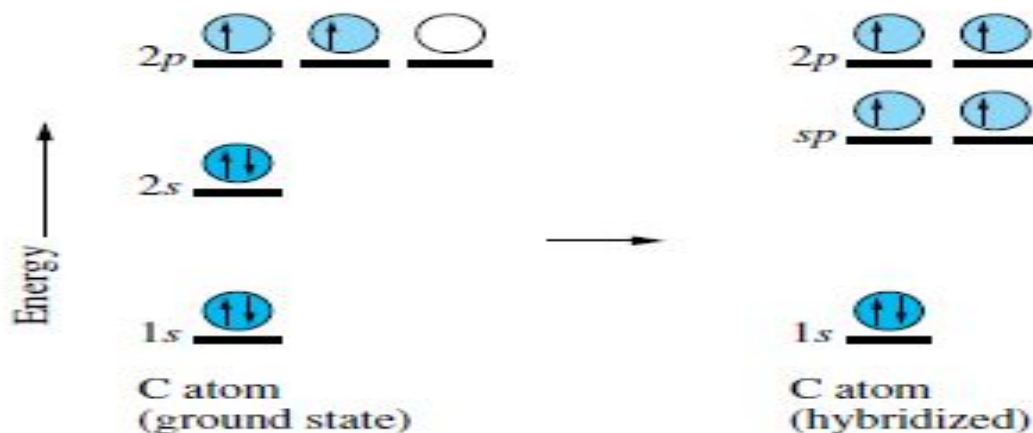
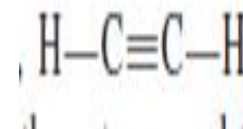


A



B

Bonding in acetylene



(A) The σ -bond framework. (B) Two 2p orbitals on each carbon atom begin to overlap (symbolized by lines) to form two π bonds.